FireSmokeGenNet: Boundary-Aware Diffusion Learning with Multimodal Quality Assessment for Wildfire Smoke Detection

Ammar Amjad, Hsien-Tsung Chang, and Li-Chia Tai

***Abstract*—Conditional diffusion models commonly treat spatial masks as deterministic constraints, an assumption that is poorly suited to wildfire smoke because its semi-transparent boundaries are intrinsically uncertain. This limitation can produce abrupt transitions, background distortion, and unrealistic object–context interactions. We address this problem through FireSmokeGenNet, a controllable latent-diffusion framework for synthetic wildfire-smoke generation and downstream detector augmentation. The framework employs a dual-branch conditioning architecture that separately encodes mask geometry and masked-image context before integrating them through joint cross-attention. Descriptive vision–language conditioning is additionally used to represent scene and smoke characteristics. We formulate Mask Random Difference Loss (MRDL), a boundary-restricted stochastic consistency objective that penalizes changes in denoising predictions under small morphological perturbations of an annotated smoke mask. Candidate samples are subsequently assessed through a task-specific quality-ranking procedure before detector training. Across five seed-matched runs, augmenting the real training set with quality-filtered FireSmokeGenNet images increased YOLOv13 AP_{50} from $78.16 \pm 0.27\%$ to $80.88 \pm 0.70\%$, an absolute improvement of 2.72 percentage points. When evaluated exclusively over the preserved region outside the conditioning smoke mask, the framework achieved a PSNR of 28.10 dB and an SSIM of 0.88. These results indicate that stochastic boundary-aware diffusion conditioning can improve synthetic-data utility for wildfire-smoke detection under the evaluated datasets, detector configurations, and experimental protocol.**

***Index Terms*—Diffusion models, Earth observation, image inpainting, remote sensing, smoke detection, synthetic data, vision-language models, wildfire monitoring.**

I. Introduction

WILDFIRES cause extensive environmental, economic, and societal damage, with severe fire seasons affecting millions of hectares and releasing substantial quantities of carbon into the atmosphere [1]–[3]. Therefore, timely wildfire detection is important for early assessment, containment planning, and emergency response. Smoke is frequently observable before flames become clearly visible and is consequently an important visual indicator for early wildfire detection [4], [5]. Unlike flames, which may be partially occluded by terrain or vegetation, smoke can rise above the forest canopy and remain visible over long distances. Vision-based smoke detection has thus become an important component of wildfire monitoring systems, complementing satellite observations, unmanned aerial vehicle surveillance, and ground-based sensor networks [5], [6].

Deep object detectors, including YOLO, DETR, and EfficientDet, have substantially advanced real-time object localization [7]–[9]. However, their effectiveness depends strongly on the availability and diversity of annotated training data. These requirements are difficult to satisfy for early wildfire smoke detection because actual wildfire events are geographically dispersed, temporally unpredictable, and hazardous to observe at close range. Existing datasets provide valuable imagery for detection and segmentation research but may represent a restricted range of fire stages, camera viewpoints, illumination conditions, seasons and weather patterns [10]. Early stage smoke is especially difficult to collect and annotate because it may appear as a faint, semi-transparent structure with low contrast against clouds, haze, fog, vegetation, or a bright sky. Models trained on restricted environmental distributions may exhibit reduced robustness under previously unseen conditions, consistent with the broader challenge of out-of-distribution generalization [11].

Synthetic data generation provides a scalable approach for increasing the diversity of smoke appearances and environmental conditions represented during detector training. Generating convincing smoke, however, differs substantially from the synthesis of rigid objects. Smoke has no fixed geometry, contains structures at multiple spatial scales, and exhibits spatially varying opacity. Its appearance depends on illumination, background texture, atmospheric conditions, and plume density [12]. Moreover, its boundaries are naturally diffuse and irregular. Therefore, a useful synthetic smoke generator must preserve the environmental context while producing diverse smoke structures, plausible translucency, and smooth transitions around uncertain plume boundaries [13].

Early synthetic smoke methods relied on computational simulations, procedural generation, and image compositing [14], [15]. Such approaches offer explicit control over smoke location and appearance but may require substantial parameter adjustments and can retain a visual gap between simulated and real smoke. GAN-based methods have also been investigated for wildfire data augmentation and controllable smoke generation [16], [17]. Although GANs offer comparatively efficient

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inference, their adversarial optimization may be affected by training instability and insufficient mode coverage, particularly when modeling fine turbulent structures and spatially varying opacities.

Diffusion models have recently achieved excellent performance in high-fidelity image synthesis. Denoising diffusion probabilistic models learn to reconstruct data by reversing a progressive Gaussian corruption process [18], whereas improved formulations introduce more effective noise schedules and reverse-process variance parameterizations [19]. Latent diffusion models reduce computational requirements by performing the denoising process in a compressed representation space and incorporating external conditions using cross-attention [20]. Spatial control architectures, such as ControlNet and T2I-Adapter, further demonstrate that external structural signals can be injected into multiple stages of a pretrained diffusion model [21], [22]. For wildfire imagery, FlameDiffuser employs mask-guided latent diffusion to generate synthetic fire scenes for data augmentation [23].

Despite this progress, three limitations remain in diffusion-based wildfire-smoke synthesis. First, standard inpainting commonly combines a noisy latent binary mask and masked image representation through input-level concatenation. This formulation does not explicitly separate the structural information from the contextual appearance. Second, deterministic binary masks provide rigid spatial constraints that do not represent the uncertain and semitransparent transitions observed at smoke boundaries. Third, diffusion sampling produces samples with varying perceptual and task-level qualities, and indiscriminately adding all generated images to detector training can introduce unrealistic colors, insufficient visibility, or implausible opacity [24]. Existing wildfire-synthesis pipelines have not jointly addressed these limitations through decoupled multiscale conditioning, boundary-specific consistency regularization, and task-aware synthetic sample filtering.

To address these limitations, we propose FireSmokeGenNet, a controllable latent-diffusion framework for the synthetic generation of wildfire smoke. The framework separately encodes the prescribed smoke mask and masked-image context, hierarchically injects their multiscale representations into the diffusion U-Net, and fuses them through Joint Cross-Attention (JCA). We further introduce the Mask Random Difference Loss (MRDL), a stochastic consistency objective that reduces the model's sensitivity to morphologically perturbed smoke boundaries. Finally, a task-specific vision-language quality assessment module ranks the generated samples according to smoke color fidelity, visibility, and translucency before they are used for downstream detector training. The work is framed as an AI-methodology contribution to controllable generation under uncertain spatial supervision. The main contributions of this work are summarized as follows:

- We propose a hierarchical dual-branch conditioning architecture that represents smoke-mask geometry and masked-image context through separate feature streams. The proposed JCA module independently attends to the structural and contextual features before fusing their representations at intermediate U-Net resolutions.
- We formulate MRDL, a boundary-restricted consistency regularizer inspired by prediction consistency under semantics-preserving perturbations [25]. MRDL penalizes differences between denoising predictions conditioned on the original and morphologically perturbed masks within the affected latent-space boundary region.
- We developed a task-specific multimodal quality-filtering strategy based on Qwen2-VL-7B [26]. Motivated by learned human preference assessment [27], [28], the module evaluates the generated smoke based on color fidelity, visibility, and translucency realism and retains the highest-ranked samples for downstream detector training.
- We establish a controlled detector-evaluation protocol comparing real-only training with training augmented by quality-filtered synthetic data across eight YOLO detector families. The evaluation additionally examines environmental shifts and fixed-seed sensitivity to the retained synthetic-sample proportion.

From an AI-methodology perspective, the central contribution is the treatment of uncertain spatial conditions as stochastic neighborhoods rather than rigid binary constraints, together with explicit structural-contextual interaction and learned multimodal sample selection. Wildfire smoke provides the evaluated application domain; transfer to other uncertain-boundary tasks is not claimed without further experiments.

II. Related Work

A. Wildfire-Smoke Datasets and Data Augmentation

The development of reliable wildfire smoke detectors is closely related to the availability and diversity of annotated data. The FLAME dataset provides aerial visible and thermal imagery acquired during prescribed pile burns and supports fire classification and segmentation [10]. Although FLAME is an important benchmark for aerial fire analysis, its acquisition under a limited set of controlled burning conditions restricts its representation of environmental and geographical variations [29]. For early smoke detection using fixed-view cameras, Dewangan *et al.* introduced FIgLib, a collection of approximately 25,000 labeled images derived from wildfire camera sequences in Southern California, together with the spatiotemporal SmokeyNet architecture [30]. The dataset includes smoke and non-smoke observations captured before and after the reported fire ignition. Nevertheless, its geographical concentration and reliance on fixed camera infrastructure may limit the direct generalization to other regions, camera configurations, and atmospheric conditions [31]. Several datasets have also been developed for smoke localization and segmentation. Smoke100K contains synthetic smoke images paired with smoke-free backgrounds, smoke masks, and bounding box annotations [32]. SMOKE5K combines real and synthetic images with pixel-level smoke annotations and explicitly considers the uncertainty associated with transparent and irregular smoke regions [33]. These resources are valuable for detection and segmentation; however, synthetic subsets constructed through compositing can exhibit appearance differences from real smoke, particularly around semi-transparent boundaries. Consequently, data augmentation has become an important strategy for improving the robustness of smoke detectors. Conventional transformations, including geometric, photometric, and color space augmentations, improve sample diversity but cannot introduce genuinely new smoke structures [34]. Compositing-based approaches offer greater control by blending foreground smoke masks with smoke-free scenes [15]. However, their realism depends strongly on the quality of the source masks, opacity estimation, color adjustment, and boundary blending. These limitations motivate generative approaches that learn the appearance of smoke directly from data.

B. Generative Models for Smoke and Wildfire Synthesis

GAN-based generation has been investigated as an alternative to procedural simulations and deterministic compositing. Park *et al.* employed CycleGAN-based augmentation to improve wildfire classification from remote camera imagery [16]. Xie and Tao proposed controllable smoke-generation networks that decompose an image into smoke and non-smoke components, allowing smoke characteristics to be modified through latent representations [17]. These studies demonstrated the potential of learned generation for expanding smoke-related data sets.

Despite their efficient inference, GAN-based methods are sensitive to adversarial training instability and insufficient mode coverage. For smoke synthesis, these issues may appear as repetitive plume structures, inconsistent opacity, or inadequate integration with the background. Moreover, methods designed for image-level augmentation do not necessarily preserve the spatial correspondence required for automatically transferring masks or bounding-box annotations to generated samples. Diffusion models provide an alternative generative formulation based on iterative denoising. DDPMs learn a reverse process that reconstructs samples from progressively corrupted observations [18]. Subsequent developments improved noise scheduling, variance parameterization, and sampling efficiency [19]. Latent diffusion models perform the denoising process within a compressed representation space and incorporate semantic conditions using cross-attention [20]. Compared with adversarial generators, diffusion models generally provide stable training and strong sample diversity, although their iterative inference incurs greater computational costs.

C. Spatially Controlled Diffusion and Image Inpainting

Diffusion-based image inpainting uses masks, image context, text, or other structural signals to constrain the denoising process. RePaint conditions a pretrained unconditional diffusion model by repeatedly resampling the known image regions during reverse diffusion, enabling inpainting across different mask configurations without mask-specific model retraining [35]. SmartBrush incorporates text and object-shape guidance to improve spatial controllability and background preservation during object inpainting [36]. InstructPix2Pix learns instruction-guided image editing from synthetically constructed editing pairs [37].

More general spatial control architectures inject external conditions into pretrained latent diffusion models. ControlNet introduces trainable control branches connected to a frozen diffusion backbone, allowing conditions such as edges, depth maps, poses, and segmentation maps to guide generation [21]. The T2I-Adapter uses lightweight adapters to align external control signals with the internal representations of a pretrained text-to-image model [22]. These methods demonstrate the value of hierarchical structural guidance, but are designed for general image synthesis rather than the uncertain, semi-transparent boundaries characteristic of smoke. For wildfire synthesis, FLAME Diffuser introduced a training-free, mask-guided diffusion pipeline that uses wildfire masks and Perlin-noise-based guidance to control the location and extent of generated fire regions [23]. This demonstrates that diffusion-generated wildfire imagery can support downstream data-augmentation tasks. However, its primary emphasis is on flame generation and paired spatial annotation rather than the explicit modeling of semi-transparent smoke interfaces.

In contrast, the present study specifically focused on smoke synthesis. It separates the mask geometry from the masked-background context through two conditioning branches, fuses the resulting representations using Joint Cross-Attention, and introduces a boundary-restricted consistency loss for stochastic mask perturbations. This formulation was designed to reduce sensitivity to smoke boundary uncertainty while retaining spatial control and environmental context.

D. Learned Assessment of Generated-Image Quality

Generated-image evaluation has traditionally relied on distributional and perceptual metrics such as Fréchet Inception Distance, structural similarity, learned perceptual similarity, and text-image embedding similarity. Although these measures capture complementary aspects of image quality, they may not reflect task-specific requirements or human judgments of realism [38].

Recent preference-based models directly learn quality assessments from human comparisons. ImageReward uses expert-ranked text–image pairs to learn a reward model for evaluating and improving text-to-image generation [27]. The Human Preference Score similarly adapts vision–language representations to predict human preferences among generated images [28]. These approaches demonstrate that learned multimodal scoring can complement conventional, automatic metrics.

Large vision-language models provide a mechanism for evaluating images according to explicitly defined semantic criteria. Qwen2-VL supports joint visual and textual reasoning across variable image resolutions [26]. However, general-purpose preference models are not specifically trained to assess the color fidelity, visibility, and translucency of smoke from wildfires. Moreover, a generated image that appears perceptually appealing is not necessarily useful for downstream smoke detection. Therefore, the proposed framework adapts a vision-language model using smoke-specific human annotations and evaluates the association between its filtering decisions and downstream wildfire-smoke detection performance through a retention-threshold sensitivity analysis.

III. Experiment

A. Preliminaries and Problem Formulation

1) Latent Diffusion Models: Latent diffusion models (LDMs) perform the diffusion process in a compressed representation space learned by a variational autoencoder (VAE), thereby reducing the computational cost of high-resolution image generation [20]. Given an RGB image

$$x \in \mathbb{R}^{H\times W\times 3}, \tag{1}$$

the VAE encoder $\mathcal{E}$ maps x to a latent representation

$$z_0 = \mathcal{E}(x) \in \mathbb{R}^{h\times w\times c}, \tag{2}$$

where $h = H/f$, $w = W/f$, and f is the spatial compression factor. For the VAE adopted in this study, $f = 8$ and $c = 4$. After reverse diffusion, the VAE decoder $\mathcal{D}$ reconstructs the output image from the denoised latent:

$$\widehat{x} = \mathcal{D}(\widehat{z}_0). \tag{3}$$

The VAE parameters remain fixed during the training of the conditional diffusion model.

2) Forward and Reverse Diffusion Processes: The diffusion formulation comprises a fixed forward corruption process and a learned reverse-denoising process. As illustrated in Fig. 1, the forward process progressively perturbs the clean latent z_0 with Gaussian noise. The reverse process predicts and removes the injected noise using the supplied conditioning information.

3) Forward Diffusion Process: Let $\{\beta_t\}_{t=1}^{T}$ denote a predefined variance schedule, where $0 < \beta_t < 1$. We define

$$\alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{\tau=1}^{t} \alpha_\tau. \tag{4}$$

The forward Markov transition is

$$q(z_t \mid z_{t-1}) = \mathcal{N}\left(z_t; \sqrt{\alpha_t}\, z_{t-1}, \beta_t I\right). \tag{5}$$

Using the reparameterization property of Gaussian distributions, z_t can be sampled directly from z_0:

$$z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{6}$$

We employed the cosine noise schedule introduced by Nichol and Dhariwal [19]. Let

$$g(t) = \cos^2\left(\frac{t/T + s_0}{1+s_0}\,\frac{\pi}{2}\right), \qquad s_0 = 0.008. \tag{7}$$

The cumulative noise schedule is normalized as

$$\bar{\alpha}_t = \frac{g(t)}{g(0)}. \tag{8}$$

The corresponding variance coefficients are obtained from successive values of $\bar{\alpha}_t$:

$$\beta_t = \min\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \beta_{\max}\right), \tag{9}$$

where $\beta_{\max} < 1$ prevents numerical instability near the final diffusion steps.

4) Conditional Reverse Diffusion Process: The reverse process reconstructs the clean latent by iteratively sampling

$$p_\theta(z_{t-1} \mid z_t, C) = \mathcal{N}\left(z_{t-1}; \mu_\theta(z_t, t, C), \sigma_t^2 I\right), \tag{10}$$

where C denotes the conditioning information and θ represents the trainable parameters of the diffusion U-Net. The diffusion timestep t is provided separately through a timestep embedding and is therefore not included in C.

Under the noise-prediction parameterization, the reverse-process mean is expressed as

$$\mu_\theta(z_t, t, C) = \frac{1}{\sqrt{\alpha_t}}\left(z_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(z_t, t, C)\right). \tag{11}$$

where $\epsilon_\theta(z_t, t, C)$ is the noise predicted by the conditional U-Net. Depending on the adopted DDPM parameterization, the reverse variance σ_t^2 may be fixed according to the posterior variance or learned during training [19]. The specific parameterization used in FireSmokeGenNet is reported in Section III-B.

5) Classifier-Free Guidance: Classifier-free guidance combines conditional and unconditional noise predictions without requiring an external classifier [39]. During training, the conditioning information is randomly replaced by a null condition. At inference time, the guided noise prediction is

$$\begin{aligned}\widetilde{\epsilon}_\theta(z_t, t, C) &= \epsilon_\theta(z_t, t, \varnothing) \\ &\quad + \gamma\left[\epsilon_\theta(z_t, t, C) - \epsilon_\theta(z_t, t, \varnothing)\right],\end{aligned} \tag{12}$$

where $\gamma \geq 1$ is the guidance scale and $\varnothing$ denotes the null condition. Increasing γ generally strengthens adherence to the conditioning information but may reduce sample diversity or introduce visual artifacts.

6) Diffusion Training Objective: The standard noise-prediction objective is

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\substack{x\sim p_{\text{data}},\, t\sim\mathcal{U}\{1,\ldots,T\},\\ \epsilon\sim\mathcal{N}(0,I)}}\left[\left\|\epsilon - \epsilon_\theta(z_t, t, C)\right\|_2^2\right], \tag{13}$$

where $z_0 = \mathcal{E}(x)$ and z_t is generated using Eq. (6). This simplified objective corresponds to a reweighted form of the variational objective and is used to optimize the reverse diffusion process. [18].

7) Conditional Score Interpretation: The noise-prediction network is related to the conditional score of the perturbed latent distribution:

$$\nabla_{z_t} \log p(z_t \mid C) \approx -\frac{\epsilon_\theta(z_t, t, C)}{\sqrt{1-\bar{\alpha}_t}}. \tag{14}$$

This interpretation connects denoising diffusion models with score-based generative modeling [40]. The conditioning information therefore guides the estimated score toward latent configurations that are compatible with the specified generation constraints.

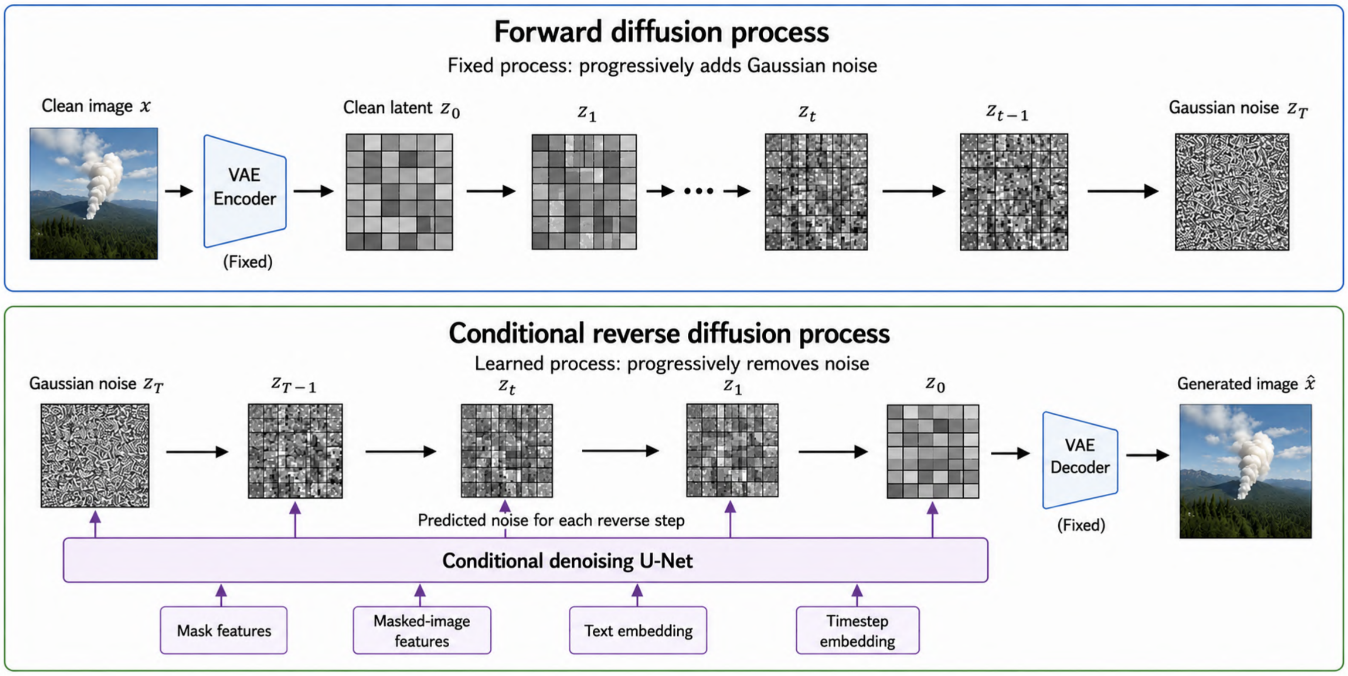

Fig. 1. **Forward and conditional reverse latent-diffusion processes.**

8) Conventional Latent-Inpainting Conditioning: Let $M \in \{0,1\}^{H\times W}$ denote a binary mask, where $M_{ij} = 1$ identifies the region to be synthesized. The corresponding masked image is

$$x_m = x \odot (1 - M), \tag{15}$$

where $\odot$ denotes elementwise multiplication. The masked image latent is obtained as

$$z_m = \mathcal{E}(x_m) \in \mathbb{R}^{h\times w\times c}. \tag{16}$$

The mask is resized to the latent spatial resolution:

$$M_z = \mathrm{Resize}(M, h, w) \in \{0,1\}^{h\times w\times 1}. \tag{17}$$

A conventional latent-inpainting U-Net forms its input through channelwise concatenation:

$$u_t = [z_t; z_m; M_z] \in \mathbb{R}^{h\times w\times (2c+1)}. \tag{18}$$

For $c = 4$, this produces a nine-channel input. This formulation is computationally convenient and provides direct spatial conditioning. However, after input-level concatenation, it does not explicitly retain separate structural and contextual conditioning pathways.

9) Limitations of Flat Conditioning for Smoke Synthesis: Although input-level concatenation is effective for general image inpainting, smoke synthesis introduces several additional requirements.

10) Implicit Multiscale Conditioning: The binary mask and masked-image latent are directly available at the input layer, whereas deeper U-Net blocks receive them through progressively transformed intermediate representations. The architecture therefore does not provide explicit multiscale access to the original structural and contextual features. This may make it more difficult to preserve global smoke geometry while refining local texture, opacity, and boundary transitions.

11) Heterogeneous Conditioning Sources: The noisy latent z_t, masked-image latent z_m, and binary mask M_z represent different types of information. The first two are continuous latent representations, whereas the mask is a discrete spatial constraint. Direct concatenation requires the initial U-Net layers to learn how to separate and integrate these heterogeneous signals without modality-specific encoders.

This does not imply that concatenation is intrinsically ineffective. Rather, it motivates investigating whether separate structural and contextual encoders can provide more explicit conditioning for smoke generation.

12) Deterministic Boundary Representation: A binary mask defines a sharp distinction between the synthesis and preservation regions. Real smoke boundaries, however, are often semi-transparent, irregular, and difficult to localize precisely. Treating a single binary contour as an exact boundary may therefore introduce an overly rigid spatial assumption. Depending on the learned behavior, this can produce abrupt transitions near the mask boundary or excessive smoke leakage outside the region.

13) Static Input-Level Representation: The mask and masked-image inputs remain fixed throughout the reverse process. Although timestep embeddings allow the U-Net to learn noise-level-dependent behavior, conventional concatenation does not explicitly control how structural and contextual information is distributed across different U-Net resolutions. This observation motivates the hierarchical conditioning strategy introduced in Section III-B.

14) Stochastic Smoke-Boundary Formulation: For smoke synthesis, the binary mask M is treated as an approximate spatial constraint rather than an exact physical boundary. We define a conditional perturbation distribution.

$$M' \sim p_{\mathcal{T}}(M' \mid M), \tag{19}$$

where $p_{\mathcal{T}}(M' \mid M)$ represents local morphological variations of M, such as controlled dilation and erosion. These perturbations do not simulate the physical evolution of smoke. Instead, they represent uncertainty in the position of the specified or annotated smoke boundary.

Let $C(M)$ and $C(M')$ denote conditioning representations obtained from the original and perturbed masks, respectively. For the same noisy latent z_t, noise realization, and timestep t, the sensitivity of the noise predictor to the mask perturbation can be expressed as

$$\Delta_\theta(M, M') = \epsilon_\theta(z_t, t, C(M)) - \epsilon_\theta(z_t, t, C(M')). \tag{20}$$

A general boundary-consistency objective can then be written as

$$\begin{aligned}\mathcal{R}_{\text{boundary}} = \mathbb{E}_{M' \sim p_{\mathcal{T}}(M'|M)} \Big[d_B \big(\epsilon_\theta(z_t, t, C(M)), \\ \epsilon_\theta(z_t, t, C(M')) \big) \Big],\end{aligned} \tag{21}$$

where $d_B(\cdot, \cdot)$ measures prediction differences within the boundary region affected by the perturbation. This expectation can be estimated through stochastic mask sampling during training.

Equation (21) defines a consistency objective over plausible mask perturbations; it is not a direct likelihood marginalization over physical smoke boundaries. The proposed MRDL provides a boundary-restricted and normalized instantiation of $d_B(\cdot, \cdot)$, as described in Section III-B.

B. Methodology

1) Overall Architecture: FireSmokeGenNet extends the SD-2 Inpainting U-Net [20] with three major modifications: (1) a dual-branch conditioning encoder, (2) JCA for feature fusion, and (3) stochastic mask perturbation via MRDL. Figure 2 illustrates the complete pipeline.

Unlike the conventional inpainting formulation, FireSmokeGenNet uses z_t as the diffusion input and injects mask and masked-image information through the proposed multiscale encoders and JCA modules.

2) Dual-Branch Conditioning Encoder: To effectively capture complementary structural and contextual information, the proposed framework employs a dual-branch conditioning encoder, as illustrated in Fig. 3. The binary smoke mask and masked background image are processed independently using dedicated backbone networks. A lightweight ResNet-18 encoder is used in the mask branch to efficiently extract structural and boundary information from the binary smoke mask. In contrast, a ResNet-50 encoder is used for the masked RGB image to capture rich semantic and contextual representations. The extracted multiscale features are independently projected to the corresponding U-Net channel dimensions. The mask and masked-image feature streams remain separate and are fused through JCA at the designated U-Net resolutions.

a) Feature Extraction: Given a binary smoke mask M and its corresponding masked image x_m, the mask branch extracts structural representations using a frozen ResNet-18 backbone, while the image branch employs a frozen ResNet-50 backbone to preserve rich semantic information learned from ImageNet pretraining. Multi-scale feature maps are extracted from both branches to facilitate hierarchical conditioning of the diffusion U-Net.

For the image branch, the extracted feature maps are

$$F_I^{(1)} \in \mathbb{R}^{H/2 \times W/2 \times 64}, \tag{22}$$

$$F_I^{(2)} \in \mathbb{R}^{H/4 \times W/4 \times 256}, \tag{23}$$

$$F_I^{(3)} \in \mathbb{R}^{H/8 \times W/8 \times 512}, \tag{24}$$

$$F_I^{(4)} \in \mathbb{R}^{H/16 \times W/16 \times 1024}, \tag{25}$$

$$F_I^{(5)} \in \mathbb{R}^{H/32 \times W/32 \times 2048}, \tag{26}$$

whereas the mask branch generates corresponding multi-scale representations

$$F_M^{(s)}, \quad s \in \{2, 3, 4, 5\}, \tag{27}$$

which capture the geometric structure of the smoke regions.

b) Feature Projection: Since the extracted feature dimensions differ from those required by the diffusion U-Net, each feature map is projected into the corresponding latent dimension using lightweight learnable 1×1 convolutional layers,

$$\hat{F}^{(s)} = \mathrm{Proj}^{(s)} \left(F^{(s)} \right), \tag{28}$$

where

$$\hat{F}^{(s)} \in \mathbb{R}^{B \times C_{\text{U-Net}}^{(s)} \times H_s \times W_s},$$

and $C_{\text{U-Net}}^{(s)} \in \{320, 640, 1280, 1280\}$ denotes the channel dimension of the corresponding U-Net stage. These projected feature maps provide complementary structural and semantic guidance for the subsequent JCA module while keeping the backbone encoders frozen throughout training.

3) Spatially Adaptive Feature Injection: Crucially, we inject features *hierarchically* based on semantic granularity. Table I specifies the injection strategy for each U-Net resolution block.

a) Justification: This design respects the information hierarchy:

- **Low resolutions ($H/32$, $H/16$):** At these coarse scales, only mask features are needed to specify where smoke should appear. Image features would add distracting texture details at a stage where structure is being formed.
- **Mid resolutions ($H/8$):** Both mask and image features are necessary. The mask provides structural constraints,

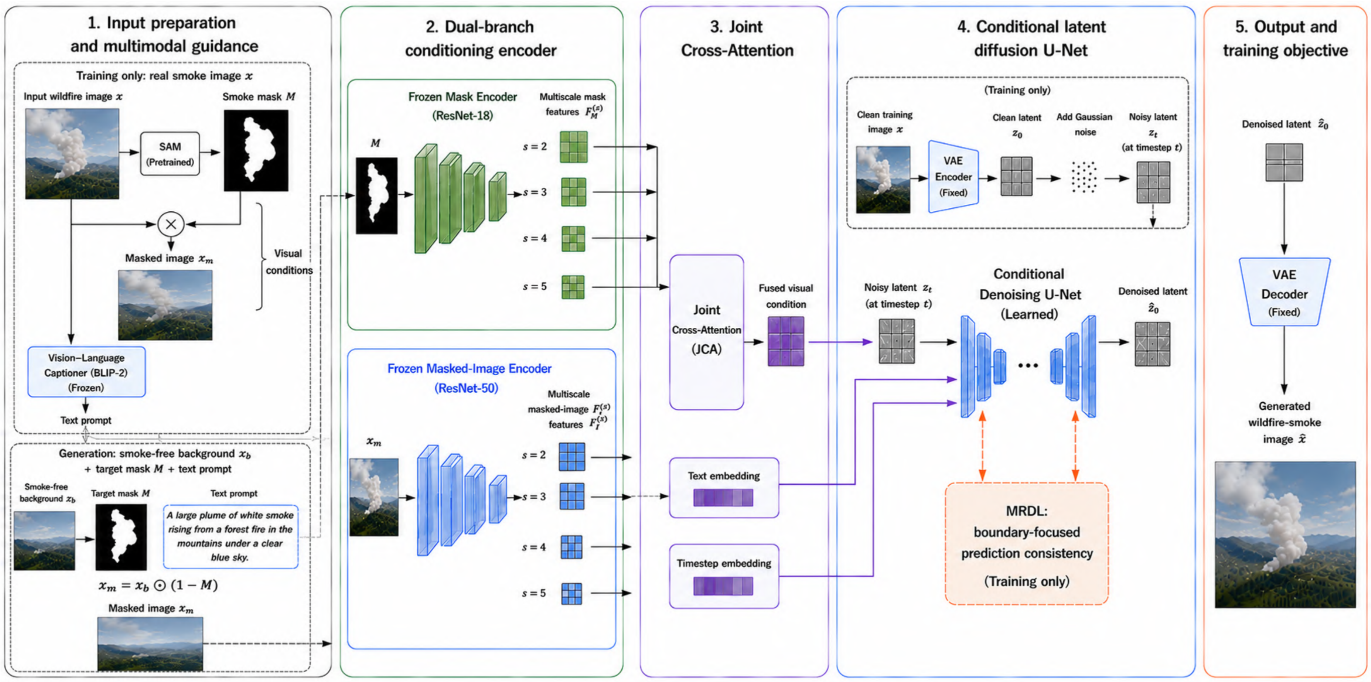

Fig. 2. **Overall architecture of FireSmokeGenNet.** During training, a real smoke image x is encoded into the clean latent z_0, while the smoke mask M, masked image x_m, text prompt, and timestep provide multimodal conditioning to the denoising U-Net. During generation, a smoke-free background x_b, target mask M, and prompt define $x_m = x_b \odot (1 - M)$; the model denoises Gaussian latent noise and decodes the result into a synthetic wildfire-smoke image.

TABLE I
SPATIALLY ADAPTIVE FEATURE INJECTION STRATEGY.

U-Net Block	Resolution	Feature Source	Injection Type
Down block 1	$H/32 \times W/32$	F_M only	Cross-attention
Down block 2	$H/16 \times W/16$	F_M only	Cross-attention
Down block 3	$H/8 \times W/8$	$F_M + F_I$	JCA
Bottleneck	$H/8 \times W/8$	F_M only	Cross-attention
Up block 1	$H/8 \times W/8$	$F_M + F_I$	JCA
Up block 2	$H/16 \times W/16$	F_I only	Cross-attention
Up block 3	$H/32 \times W/32$	F_I only	Cross-attention
Up block 4	$H/64 \times W/64$	F_I only	Cross-attention

while the image provides context for blending (e.g., matching background colors).

- **High resolutions ($H/16$, $H/32$, $H/64$):** Only image features are injected. At these fine scales, the mask is already satisfied; the model focuses on texture synthesis and boundary refinement.

4) Joint Cross-Attention (JCA): To effectively integrate structural information from the smoke mask with semantic context extracted from the masked image, we propose the JCA module, as illustrated in Fig. 4. Unlike conventional cross-attention, which relies on a single conditioning source, JCA independently models interactions between the U-Net latent features and the mask and image representations. The resulting attention features are subsequently fused to provide richer multimodal guidance for smoke synthesis.

Standard cross-attention is computed as:

$$\mathrm{CrossAttn}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{29}$$

where Q is derived from the U-Net latent features, while K and V are obtained from the conditioning features.

JCA computes separate attention for each conditioning branch and fuses the resulting representations through a lightweight multilayer perceptron (MLP):

$$Q = \mathrm{Linear}_Q(X) \in \mathbb{R}^{B \times N \times d}, \tag{30}$$
$$K_M = \mathrm{Linear}_K(\hat{F}_M), \qquad V_M = \mathrm{Linear}_V(\hat{F}_M), \tag{31}$$
$$K_I = \mathrm{Linear}_K(\hat{F}_I), \qquad V_I = \mathrm{Linear}_V(\hat{F}_I), \tag{32}$$
$$Z_M = \mathrm{CrossAttn}(Q, K_M, V_M), \tag{33}$$
$$Z_I = \mathrm{CrossAttn}(Q, K_I, V_I), \tag{34}$$
$$X' = X + \mathrm{MLP}([Z_M; Z_I]), \tag{35}$$

where $[\cdot;\cdot]$ denotes channel-wise concatenation, and the MLP consists of two fully connected layers with a hidden dimension of $4d$.

a) Computational Complexity.: For each conditioning branch, cross-attention between N U-Net tokens and L conditioning tokens requires $\mathcal{O}(NLd)$ operations, where d is the attention dimension. Because JCA applies cross-attention independently to the mask and masked-image branches, its attention cost is $\mathcal{O}(2NLd)$, excluding the linear projections and fusion MLP.

5) Mask Random Difference Loss (MRDL): MRDL addresses the deterministic boundary assumption by training the model to be invariant to plausible mask perturbations.

a) Morphological Perturbation Generation: Figure 5 illustrates the stochastic perturbation process used to represent uncertainty around the annotated smoke boundary.

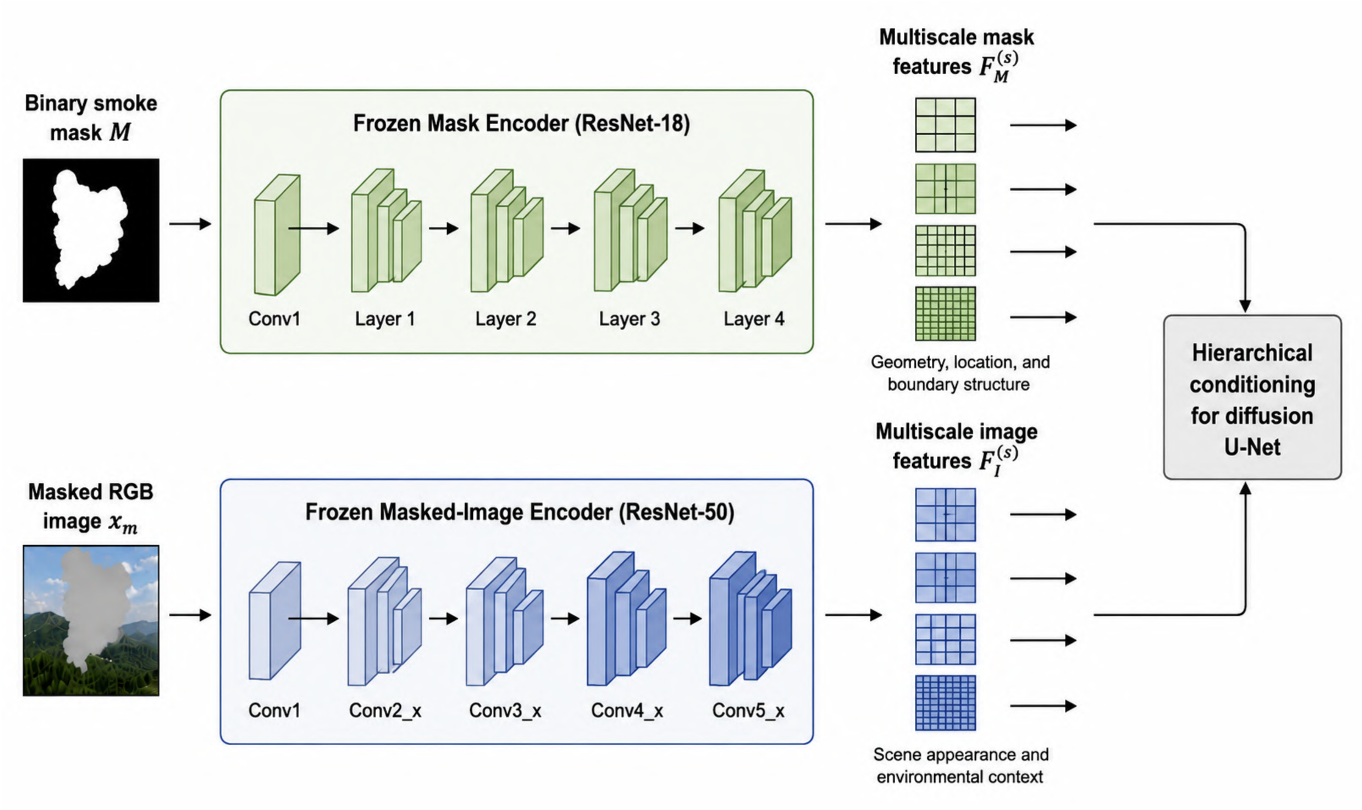

Fig. 3. **Dual-branch conditioning encoder.** The frozen ResNet-18 mask encoder extracts multiscale smoke-geometry features $F_M^{(s)}$, while the frozen ResNet-50 masked-image encoder extracts multiscale scene-context features $F_I^{(s)}$. The feature streams remain separate and are injected hierarchically into the diffusion U-Net.

For each training iteration, a perturbed mask M' is generated by applying a sequence of random morphological operations:

$$M' = \mathrm{Morph}(M, k, N), \tag{36}$$

where $k \sim \mathcal{U}_{\mathbb{Z}}(10, 20)$ denotes the structuring-element radius and $N \sim \mathcal{U}_{\mathbb{Z}}(1, 3)$ denotes the number of consecutive operations. At each perturbation step, dilation or erosion is selected with equal probability. The corresponding square structuring element has a side length $2k + 1$ pixels.

Using a square structuring element, dilation is defined as

$$\left[\mathrm{Dilate}(M, k)\right]_{ij} = \max_{\substack{-k \le u \le k \\ -k \le v \le k}} M_{i+u, j+v}. \tag{37}$$

Similarly, erosion is defined as

$$\left[\mathrm{Erode}(M, k)\right]_{ij} = \min_{\substack{-k \le u \le k \\ -k \le v \le k}} M_{i+u, j+v}. \tag{38}$$

b) Loss Formulation: For an original mask M and its perturbed realization M', we define the changed boundary region by the symmetric difference

$$B(M, M') = \mathrm{Resize}(|M - M'|, h, w), \tag{39}$$

where nearest-neighbor interpolation maps the binary region to latent resolution. The same clean latent z_0, Gaussian noise ϵ, and timestep t are used to construct z_t for both conditions. MRDL then compares the two denoising predictions only inside the affected region:

$$\mathcal{L}_{\mathrm{MRDL}} = \mathbb{E}_{M',t,\epsilon}\left[\frac{\|B \odot (\widehat{\epsilon}_M - \mathrm{sg}[\widehat{\epsilon}_{M'}])\|_2^2}{\|B\|_1 + \eta}\right], \quad B = B(M, M'). \tag{40}$$

where $\mathrm{sg}[\cdot]$ denotes stop-gradient and $\eta = 10^{-6}$ prevents division by zero. Detaching the perturbed mask prediction defines a one-sided consistency target and prevents both branches from moving simultaneously toward a trivial agreement solution; this is an optimization convention rather than an assertion that the perturbed mask is intrinsically more accurate. Samples producing an empty changed region are resampled. This normalized consistency loss prevents large masks from dominating the minibatch objective.

c) Regularization Interpretation: MRDL is a local consistency regularizer over plausible annotation-boundary perturbations. It reduces the sensitivity of the conditional score estimate to small changes in the supplied binary contour; it is not an exact marginalization of the data likelihood over physical smoke boundaries.

d) Total Training Loss: The complete training objective combines the standard diffusion loss with the proposed MRDL:

$$\mathcal{L}_{\mathrm{total}} = (1 - \omega)\mathcal{L}_{\mathrm{diff}} + \omega\mathcal{L}_{\mathrm{MRDL}}, \tag{41}$$

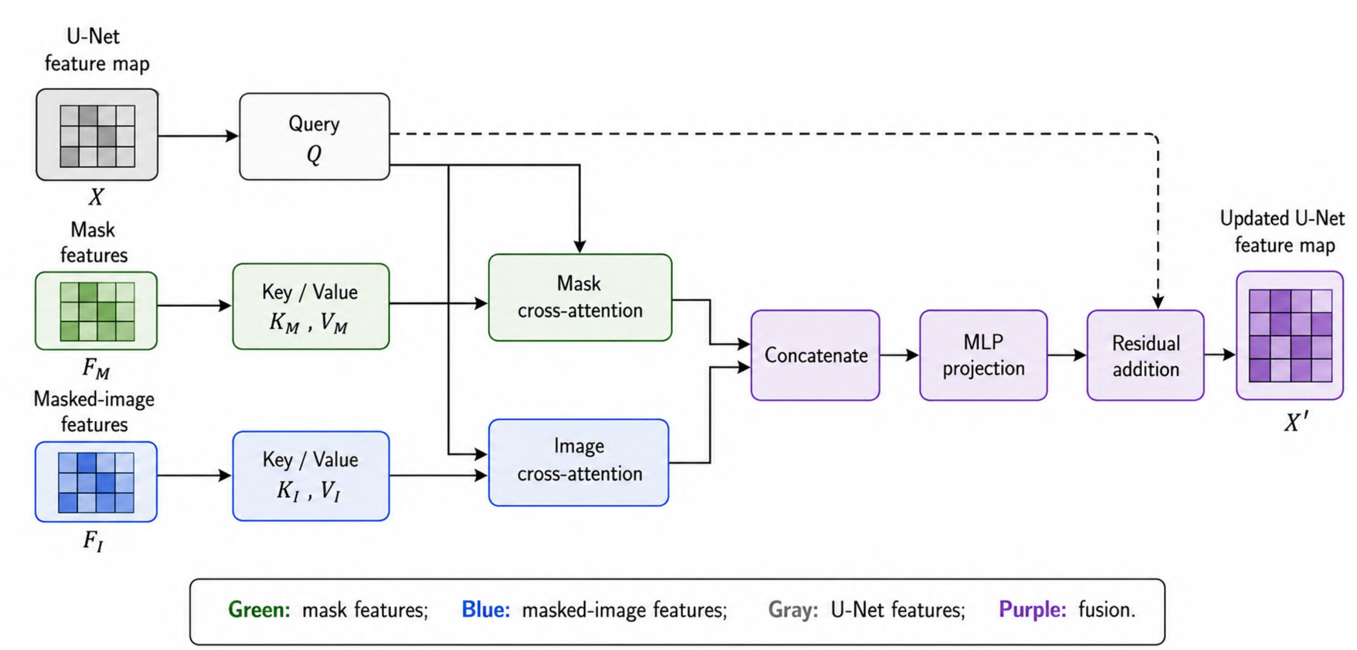

Fig. 4. **Joint Cross-Attention (JCA) module.**

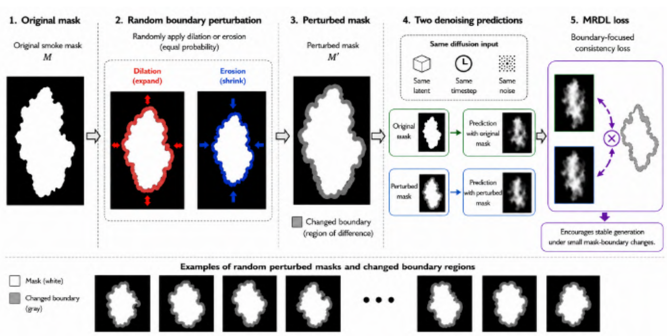

Fig. 5. **Mask Random Difference Loss (MRDL).** A smoke mask is randomly perturbed by dilation or erosion to obtain a perturbed mask and its changed boundary region. Denoising predictions conditioned on the original and perturbed masks are compared only within this region, encouraging stable generation under small mask-boundary changes.

where ω controls the strength of the boundary-consistency regularization. Based on the ablation study in Section VI-F, we set $\omega = 0.4$ in all reported experiments.

6) Regularization Analysis of MRDL: Let $f_\theta(M) = \epsilon_\theta(z_t, t, C(M))$ for fixed z_t and t. For sufficiently small contour perturbations $\Delta M = M' - M$, a first-order expansion gives

$$f_\theta(M') - f_\theta(M) \approx J_{f_\theta}(M)\Delta M, \tag{42}$$

where $J_{f_\theta}(M)$ is the Jacobian of the conditional prediction with respect to the mask condition. Minimizing Eq. (40) therefore penalizes the locally weighted response $B(M, M') \odot J_{f_\theta}(M)\Delta M$. This encourages local stability around uncertain mask contours while leaving predictions away from the perturbed boundary unconstrained. The analysis motivates MRDL as a Jacobian-style consistency penalty and does not assume that morphological perturbations reproduce the physical evolution of smoke.

7) Multimodal Quality Filtering: Diffusion stochasticity produces variable-quality outputs. We implement a three-stage filtering pipeline.

a) Stage 1: Manual Annotation: We randomly sample 150 generated images and manually annotate each on three axes (0-10 scale):

- **Color fidelity** (s_{color}): Alignment with natural smoke tones (grays, browns, blues). Score 10 for perfect color, 0 for completely unnatural colors (e.g., neon green).
- **Visibility** ($s_{\text{visibility}}$): Smoke prominence against background. Score 10 for clearly visible but not overwhelming, 0 for invisible or fully opaque.
- **Translucency** ($s_{\text{translucency}}$): Semi-transparency and background blending. Score 10 for a realistic transparency gradient; 0 for solid, opaque, or fully transparent.

To ensure consistency, annotators were provided with reference images and a shared rubric. Agreement for the continuous scores was quantified using a two-way random-effects, absolute-agreement intraclass correlation coefficient (ICC(2,k)); the observed agreement was 0.82. Because an uncertainty interval was not estimated, this value is interpreted descriptively. The VLM is used as a within-study ranking mechanism rather than as a generally calibrated perceptual-quality assessor.

b) Stage 2: VLM Fine-Tuning: We fine-tune Qwen2-VL-7B-Instruct [26] to predict quality scores. The model receives an image-caption pair and outputs three scores:

$$(s_{\text{color}}, s_{\text{visibility}}, s_{\text{translucency}}) = \text{Qwen2-VL}(x, c) \tag{43}$$

Fine-tuning uses LoRA [41] (rank $r = 16$) to reduce memory footprint, with learning rate 1×10^{-4} for 10 epochs.

The training loss is MSE between predicted and ground-truth scores. Because the annotation set contains 150 images, the VLM is used only as a task-specific ranking component in this pipeline; no broader generalization of its absolute scores is claimed.

c) Stage 3: Filtering: The composite quality score used fixed, manually specified weights:

$$Q = 0.4s_{\text{color}} + 0.4s_{\text{visibility}} + 0.2s_{\text{translucency}}. \tag{44}$$

These weights were specified before the downstream detector experiments and placed greater emphasis on color fidelity and smoke visibility. They were not optimized using detector-test performance. Translucency received a lower weight because its assessment was considered more subjective and less reliable within the adopted annotation protocol.

We generated 96,000 candidate samples and computed Q for each sample using the fine-tuned VLM. The candidates were ranked by Q, and the top 30% (28,800 samples) were retained for downstream detector training.

d) Bounding Box Annotation: Since we have the original mask M, we can automatically extract bounding boxes without manual labeling. For each mask, we compute:

$$B = \text{BBox}(\text{MaxComponent}(M)) \tag{45}$$

where MaxComponent selects the largest connected component (to filter out noise), and BBox returns $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$. This provides detection labels directly.

8) Distribution Consistency Validation: Real and synthetic images are compared in one common feature space. Let

$$f_r = \phi(x_r), \qquad f_s = \phi(x_s), \tag{46}$$

where ϕ denotes the CLIP ViT-H/14 image encoder used for the results in Table XIII. We report the Fréchet distance in this feature space (CLIP-FD)

$$\text{CLIP-FD} = \|\mu_r - \mu_s\|_2^2 + \text{Tr}\Big(\Sigma_r + \Sigma_s - 2(\Sigma_r\Sigma_s)^{1/2}\Big), \tag{47}$$

and linear-kernel MMD

$$\text{MMD}^2_{\text{linear}} = \left\| \frac{1}{n}\sum_i f_{r,i} - \frac{1}{m}\sum_j f_{s,j} \right\|_2^2, \tag{48}$$

Lower values indicate closer feature distributions. Both measures were computed from the same fixed real and synthetic image manifests. The results are interpreted descriptively because image-level uncertainty intervals were not estimated.

9) Computational Considerations: The proposed method adds two feature encoders and a second modality-specific cross-attention path at selected U-Net resolutions.

a) Dual Encoder: Feature extraction scales linearly with the input spatial size, $\mathcal{O}(HW)$, for a fixed encoder configuration.

TABLE II
COMPOSITION OF THE REAL AND SYNTHETIC DATA PARTITIONS. THE REAL-SMOKE AND REAL-NON-SMOKE COLUMNS EXCLUDE SYNTHETIC IMAGES.

Partition	Sources	Real smoke	Real non-smoke	Real total	Synthetic smoke	Total
Training	480	12,000	12,000	24,000	28,800	52,800
Validation	60	1,500	1,500	3,000	0	3,000
Test	60	1,500	1,500	3,000	0	3,000
Total	600	15,000	15,000	30,000	28,800	58,800

b) Joint Cross-Attention: Let N denote the number of U-Net latent tokens, L the number of conditioning tokens, and d the attention dimension. Each cross-attention branch requires $\mathcal{O}(NLd)$ operations. Because JCA applies separate cross-attention to mask and masked-image features, its attention cost is $\mathcal{O}(2NLd)$, excluding the linear projections and fusion MLP. The practical cost also depends on the feature dimensions, attention heads, and injection stages. The complete system required approximately 2 seconds to generate one 512×512 image with 50 DDIM steps on an NVIDIA A100 GPU.

c) MRDL: Morphological perturbations operate in $\mathcal{O}(HW)$ time. Their standalone cost was not reported separately from the complete training pipeline.

C. Experimental Setup

1) Datasets:

a) Dataset Composition and Partitioning: The experimental dataset comprised 30,000 real images obtained from 600 independent source scenes or videos. Before any frame, mask, or background extraction, the source scenes and videos were divided into training, validation, and test partitions using an 80:10:10 ratio. This source-level partitioning prevented frames originating from the same scene or video from appearing in more than one partition. Each real-data partition contained equal numbers of smoke and non-smoke images.

The training partition contained 24,000 real images from 480 independent sources, including 12,000 smoke and 12,000 non-smoke images. The validation and test partitions each contained 3,000 real images from 60 independent sources, comprising 1,500 smoke and 1,500 non-smoke images. An additional 28,800 synthetic smoke images were included only in the mixed-data training condition. No synthetic images were included in the validation or test partitions. Table II summarizes the resulting partitions.

b) Split and Leakage Control: All dataset partitions were established at the source-scene or source-video level before frame extraction, smoke-mask generation, or background selection. Consequently, temporally adjacent frames from the same video and images derived from the same source scene were assigned to a single partition.

Exact duplicate images were identified by comparing SHA-256 hashes. Near-duplicate images were detected using perceptual hashes with a maximum Hamming-distance threshold of 5. Source-video identifiers, source-scene identifiers, background identifiers, and mask-source identifiers were additionally compared across partitions. Any image or source association that violated the partitioning constraints was removed before model training. As summarized in Table XXI, the candidate-pool audit identified 184 exact duplicates and 627 near-duplicates; all were excluded before the final partition manifests and the counts in Table II were frozen. The final cross-partition audit found no shared scene/video, background, or mask-source identifiers.

Synthetic images were used exclusively in the training partition. The validation and test partitions contained only real images. Furthermore, the source scenes and smoke-free backgrounds employed for synthetic-image generation were excluded from the detector validation and test sets. The leakage-control procedure therefore operated at both the source level and the image-content level.

c) Smoke-Free Backgrounds: A total of 8,000 smoke-free background images with a minimum spatial resolution of 1920×1080 pixels were curated from sources whose licenses or terms of use permitted their use for research. The source composition was as follows:

- forest-surveillance cameras: 3,200 images (40%);
- public landscape datasets: 2,400 images (30%);
- aerial-drone footage: 1,600 images (20%); and
- other licensed forest-scene sources: 800 images (10%).

The backgrounds were categorized according to four environmental attributes:

- **Season:** summer, 40%; autumn, 30%; winter, 20%; and spring, 10%;
- **Illumination:** daytime, 60%; dusk, 25%; and nighttime, 15%;
- **Weather:** clear, 50%; haze, 30%; and fog, 20%; and
- **Terrain:** flat, 40%; hilly, 35%; and mountainous, 25%.

The four attributes were assigned independently; therefore, their categories were not mutually exclusive across attributes. For example, an image could simultaneously be categorized as a summer, daytime, clear-weather, and mountainous scene.

Each candidate background was manually inspected to confirm the absence of visible smoke and fire. The dataset manifest recorded the background identifier, source category, source URL or public-dataset identifier, license or applicable terms of use, collection date, and environmental attributes. These records were also used to enforce source-level separation between the synthetic training data and the real validation and test data.

d) Smoke-Mask Extraction: Candidate smoke masks were extracted from smoke-containing images in FLAME [10], HPWREN [42], and SMOKE5K [33]. For each candidate image, a bounding-box prompt enclosing the visible smoke region was provided to the Segment Anything Model (SAM) [43]. The resulting segmentation was converted into a binary smoke mask and associated with its source-image and source-dataset identifiers.

The initial extraction produced 65,000 candidate masks. Each mask was inspected using the predefined quality-control criteria of smoke-region coverage, boundary accuracy, connectedness, and correspondence with the visible smoke plume. Approximately 5,000 masks exhibiting severe segmentation errors, fragmented smoke regions, insufficient smoke coverage, or inaccurate boundaries were excluded. The final mask pool therefore contained 60,000 masks. Table III summarizes the extraction and exclusion counts.

TABLE III
SMOKE-MASK EXTRACTION AND QUALITY-CONTROL SUMMARY.

Dataset	Extracted	Excluded	Retained
FLAME	43,000	3,000	40,000
HPWREN	16,500	1,500	15,000
SMOKE5K	5,500	500	5,000
Total	65,000	5,000	60,000

TABLE IV
SYNTHETIC-DATA GENERATION AND QUALITY-FILTERING CONFIGURATION.

Item	Value
Eligible backgrounds	8,000
Available smoke masks	60,000
Masks sampled per background	3
Background-mask pairs	24,000
Stochastic variations per pair	4
Generated candidate images	96,000
VLM retention proportion	30%
Retained synthetic images	28,800

e) Synthetic-Image Generation: Each of the 8,000 eligible smoke-free backgrounds was paired with three smoke masks through stratified random sampling. Stratification was performed according to the recorded environmental attributes and mask-source dataset to avoid excessive concentration on a single background category or mask source. This procedure produced 24,000 background-mask pairs.

Four stochastic variations were generated for each background-mask pair, producing 96,000 candidate synthetic images. The four variations shared the same background and conditioning mask but used different generation seeds. For every generated image, the corresponding generation seed, background identifier, mask identifier, text condition, model checkpoint, and inference configuration were stored in the generation manifest.

The fine-tuned vision-language model assigned a composite quality score to every candidate image. The candidates were ranked according to this score, and the highest-ranked 30%, corresponding to 28,800 synthetic smoke images, were retained for mixed-data detector training. Table IV summarizes the generation and filtering procedure.

f) Environmental-Shift Subsets: Three environmental-shift experiments were constructed to evaluate detector generalization across changes in season, illumination, and weather. The seasonal experiment used summer images for training and winter images for testing. The illumination experiment used daytime images for training and dusk images for testing. The weather experiment used clear-condition images for training and haze-or-fog images for testing. Table V reports the corresponding image and source counts.

For each environmental-shift experiment, partitioning was performed at the source-scene or source-video level. No source scene, source video, background image, or near-duplicate image was shared between the corresponding training and test subsets. Images used in different environmental-shift experiments could overlap because each experiment evaluated

TABLE V
ENVIRONMENTAL-SHIFT EXPERIMENTAL SUBSETS.

Shift	Train images	Train sources	Test images	Test sources
Summer $\rightarrow$ Winter	9,600	192	3,000	60
Day $\rightarrow$ Dusk	14,400	288	3,750	75
Clear $\rightarrow$ Haze/Fog	12,000	240	6,000	120

TABLE VI
GENERATOR-TRAINING HYPERPARAMETERS.

Parameter	Value
Optimizer	AdamW
β_1, β_2	$0.9, 0.999$
Initial learning rate	1×10^{-4}
Learning-rate schedule	Cosine decay
Warm-up iterations	1,000
Weight decay	1×10^{-2}
Batch size	32
Training iterations	20,000
Numerical precision	FP16
Gradient checkpointing	Enabled
EMA decay	0.9999
MRDL weight ω	0.4

a different environmental attribute; however, the training and test subsets within an individual experiment remained mutually exclusive at the source level.

2) Implementation Details:

a) Software Environment: The implementation used Python 3.11.9, PyTorch 2.5.1, CUDA 12.4, cuDNN 9.1, and Hugging Face Diffusers 0.32.2. The exact package versions, repository commits, configuration files, and execution commands were archived after the experiments were completed.

b) Detector Implementations and Evaluation Protocol: The detector evaluation included YOLOv6-YOLOv13. YOLOv12 followed the attention-centric detector proposed by Tian *et al.* [44], whereas YOLOv13 followed the hypergraph-enhanced detector proposed by Lei *et al.* [45]. The YOLOv12 and YOLOv13 implementations were obtained from https://github.com/sunsmarterjie/yolov12 and https://github.com/iMoonLab/yolov13, respectively.

For each detector family, the real-only and mixed-data conditions used the same model size, initialization, data partitions, preprocessing, augmentation, optimizer, training duration, confidence threshold, and non-maximum suppression threshold. Only the composition of the training data differed between the paired conditions.

c) Hardware: The experimental platform consisted of one NVIDIA A100 GPU with 80 GB of memory, a 32-core Intel Xeon CPU, 256 GB of system memory, and 4 TB of NVMe storage.

d) Generator Training Hyperparameters: The diffusion generator was trained using the configuration in Table VI.

e) Detector Training Hyperparameters: All detector variants used an input resolution of 640×640. The complete detector-training configuration is reported in Table VII. Detector training used AdamW with an initial learning rate of 1×10^{-3}, a batch size of 16, and a maximum of 200 epochs.

The augmentation pipeline included horizontal flipping with a probability of 0.5, scaling within $[0.8, 1.2]$, and translation of up to 10%, and HSV color augmentation. The same augmentation parameters were used for the real-only and mixed-data conditions.

TABLE VII
DETECTOR-TRAINING AND EVALUATION CONFIGURATION.

Parameter	Value
Input resolution	640×640
Optimizer	AdamW
Initial learning rate	1×10^{-3}
Batch size	16
Maximum epochs	200
Warm-up epochs	3
Weight decay	5×10^{-4}
Early-stopping patience	30 epochs
Confidence threshold	0.25
NMS IoU threshold	0.45

TABLE VIII
SYNTHETIC-IMAGE INFERENCE HYPERPARAMETERS.

Parameter	Value
Sampling algorithm	DDIM
Sampling steps	50
Classifier-free guidance scale	7.5
Development seed	42
Generation seeds	Stored per image
Output resolution	512×512

f) Inference Hyperparameters: Synthetic images were generated using DDIM [46] with the configuration shown in Table VIII.

g) Architecture Details: The latent diffusion U-Net operated at a latent resolution of 64×64. The conditioning structure is summarized in Table IX. Here, F_m denotes mask features, F_I denotes masked-image features, and JCA denotes the proposed Joint Cross-Attention module.

h) Computational-Cost Measurement: Computational characteristics were measured at 512×512 resolution with batch size one and FP16 inference on the NVIDIA A100. To maintain a common scope, parameter counts cover the denoising U-Net and method-specific conditioning modules, excluding the shared frozen text encoder and VAE. FLOPs denote one denoising-step forward pass rather than the complete 50-step trajectory. Peak allocated GPU memory was recorded after 50 warm-up generations, and end-to-end latency was averaged over 1,000 synchronized 50-step generations. Table XXII reports these measurements. Total training time and offline VLM-scoring cost were not included.

i) Equal-Budget Generator Protocol: The downstream generator comparison used 28,800 retained synthetic images for every augmentation condition. The GAN baseline used the CycleGAN-based smoke-synthesis implementation in [16]; SD-Inpainting and FlameDiffuser used their cited public implementations. All methods used the same eligible backgrounds, source-level detector partitions, YOLOv13 model variant, detector hyperparameters, and five matched detector seeds. No synthetic image entered validation or test data. Table XVIII therefore compares training-source effects at a fixed retained-image budget.

TABLE IX
U-NET ARCHITECTURE AND CONDITIONING CONFIGURATION.

Stage	Resolution	Channels	Attention	Conditioning
Input	64×64	320	–	–
Down 1	32×32	320	–	F_M
Down 2	16×16	640	–	F_M
Down 3	8×8	1280	✓	JCA
Bottleneck	8×8	1280	✓	F_M
Up 1	8×8	1280	✓	JCA
Up 2	16×16	640	–	F_I
Up 3	32×32	320	–	F_I
Up 4	64×64	320	–	F_I
Output	64×64	320	–	–

j) Multi-Seed Ablation and Selection Protocol: For each configuration in Table XIX, the generator was trained with the schedule in Table VI while changing only the named component or objective. A fixed synthetic manifest was then generated for that configuration, and YOLOv13 was trained with the five matched seeds used in the principal detector experiment. Accordingly, the reported variation describes detector-training variability for one fixed synthetic manifest per generator configuration; it does not quantify generator-training or candidate-generation variability. The equal-size selection experiment used the same 96,000-candidate pool and retained 28,800 images per strategy. Random selection used a stored random permutation; bottom- and top-ranked selection used the task-specific VLM score; rule-based filtering applied the same mask-coverage and boundary-quality criteria without VLM ranking.

3) Repeated-Run and Statistical Protocol: All principal detector experiments were repeated using five random seeds: 42, 123, 3407, 2025, and 2026. Each seed controlled the model initialization, data ordering, stochastic augmentation, and data-loader sampling. Identical seeds were matched between the real-only and synthetic-augmentation conditions, thereby enabling paired statistical comparisons.

For a metric x measured over $n = 5$ runs, the sample mean was calculated as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \tag{49}$$

and the sample standard deviation was calculated as

$$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2}. \tag{50}$$

The two-sided 95% confidence interval for the population mean was calculated using the Student's t distribution:

$$\mathrm{CI}_{95\%} = \bar{x} \pm t_{0.975,n-1}\frac{s_x}{\sqrt{n}}, \tag{51}$$

where $t_{0.975,n-1}$ denotes the 97.5th percentile of the Student's t distribution with $n-1$ degrees of freedom.

The matched-pair hypothesis tests, multiplicity correction procedure, and effect-size definitions are specified after the metric definitions below. The individual seed-level results, sample means, sample standard deviations, 95% confidence intervals, exact adjusted p-values, and paired effect sizes are reported in Table X.

4) Evaluation Metrics:

a) Background-Preservation Metrics: PSNR, SSIM, and LPIPS required paired references. They were therefore computed over the preserved background region $1 - M$ by comparing each generated image with its source background after identical resizing and normalization to $[0, 1]$. These metrics evaluated background preservation rather than smoke realism.

- **PSNR (dB):** Peak signal-to-noise ratio; higher is better.
- **SSIM** [47]: Structural similarity; higher is better.
- **1-LPIPS** [48]: One minus learned perceptual image patch similarity; higher is better.
- **CLIP-Sim** [49]: cosine similarity between the complete generated image and its caption: higher is better.

All methods were evaluated using the same image manifest, mask manifest, resizing procedure, normalization, and metric implementations. CLIP-Sim was calculated using the CLIP ViT-H/14 image encoder.

b) Detection Metrics: Detector performance was evaluated using:

- **AP**$_{50}$: average precision at an IoU threshold of 0.5;
- **AP**$_{50:95}$: average precision over IoU thresholds from 0.5 to 0.95 in increments of 0.05;
- **Precision**: $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$;
- **Recall**: $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$.

c) Boundary-Quality Metrics: The boundary band associated with the smoke mask M was defined as

$$B_\delta(M) = \mathrm{Dilate}(M,\delta) \setminus \mathrm{Erode}(M,\delta), \qquad \delta = 5, \tag{52}$$

where δ denotes the morphological kernel radius in pixels. The boundary-softness score was calculated as

$$S(M, I) = \frac{1}{|B_\delta(M)|}\sum_{p\in B_\delta(M)} \|\nabla I(p)\|_2\,, \tag{53}$$

where I denotes the generated image and $\nabla I(p)$ is the spatial image gradient at pixel p. Lower abrupt-gradient concentrations around the smoke boundary indicate smoother integration between the generated smoke and the surrounding scene.

The divergence between the real and synthetic boundary-gradient distributions were calculated as

$$G = D_{\mathrm{KL}}\left(p_{\mathrm{real}}(g) \,\|\, p_{\mathrm{synth}}(g)\right), \tag{54}$$

where the boundary-gradient magnitude was defined as

$$g(p) = \|\nabla I(p)\|_2\,, \qquad p \in B_\delta(M). \tag{55}$$

The gradient distributions were estimated using 100-bin normalized histograms. A smoothing constant of 10^{-8} was added to every bin before calculating the KL divergence to avoid undefined logarithms caused by zero-probability bins. The same boundary width, histogram estimator, smoothing constant, and image-preprocessing procedure were applied to all compared methods. A lower KL divergence indicates that the synthetic boundary-gradient distribution more closely resembles that of real smoke images.

d) Hypothesis Testing and Effect Sizes: Statistical comparisons were performed using the paired performance differences obtained from matched seeds. The normality of the paired differences was examined using the Shapiro-Wilk test. A two-sided paired t-test was applied when the normality assumption was not rejected at $\alpha = 0.05$. When this assumption was rejected, a two-sided Wilcoxon signed-rank test was used as a sensitivity analysis.

Because eight detector-family comparisons were performed, the resulting p-values were corrected using the Holm-Bonferroni procedure. Statistical significance was determined using the adjusted p-values at $\alpha = 0.05$. For paired t-tests, the standardized paired effect size was calculated as

$$d_z = \frac{\overline{D}}{s_D}, \tag{56}$$

where $\overline{D}$ and s_D denote the mean and sample standard deviation of the paired performance differences, respectively. For a Wilcoxon comparison, matched-pairs rank-biserial correlation was used instead.

e) Statistical Results: As reported in Table X, the mixed-data condition increased the mean AP_{50} of every evaluated detector. The mean improvements ranged from 2.60 to 2.78 percentage points. The largest improvement was observed for YOLOv8, for which AP_{50} increased from $74.18 \pm 0.29\%$ to $76.96 \pm 0.50\%$.

After Holm-Bonferroni correction, all paired comparisons remained statistically significant, with adjusted p-values ranging from 0.0019 to 0.0022. The paired Cohen's d_z values ranged from 3.94 to 5.59, indicating large standardized effects under the five-run configuration. The narrow confidence intervals observed for most detector configurations also indicated limited performance variation across the five evaluated seeds.

The average improvement across the eight detector families was 2.66 percentage points. These repeated-run results indicated that the performance gains were not attributable to a single random initialization. Instead, the inclusion of quality-filtered synthetic smoke images consistently improved detection performance across the evaluated YOLO architectures.

Nevertheless, because each condition contained only five repeated runs, the statistical findings were interpreted together with their confidence intervals and effect sizes rather than solely through significance testing. The results support the observed utility of the proposed synthetic-data pipeline under the evaluated datasets, detector configurations, and experimental protocol.

f) Human and VLM Quality Assessment: A total of 150 generated images was independently rated by three annotators with experience in computer vision and wildfire-image analysis. Every annotator assessed every image for color fidelity, smoke visibility, and translucency using a 0-10 scale.

Inter-rater agreement was quantified using a two-way random-effects, absolute-agreement intraclass correlation coefficient, ICC(2,k). The observed agreement was 0.82. Because an uncertainty interval was not estimated, the value is interpreted descriptively. Because the annotation set was small, the fine-tuned VLM was treated only as a within-study ranking model; no claim of general score calibration or transfer beyond the evaluated candidate pool was made. Predictive agreement was evaluated using five-fold image-level cross-validation, such that the prediction for each image was produced by a model that had not been fine-tuned on that image. The folds were not designed as source-grouped cross-validation; consequently, the reported values assess within-pool ranking and are not interpreted as evidence of transfer across backgrounds, source scenes, or datasets. No external calibration dataset was available.

IV. Results

A. Vision-Language Quality-Predictor Validation

The task-specific VLM was evaluated using five-fold image-level cross-validation on 150 human-rated images. In each fold, 120 images were used for LoRA fine-tuning and 30 unseen images were used for evaluation. Each fold was initialized independently from the same Qwen2-VL-7B-Instruct checkpoint.

Agreement with the mean human ratings was measured using MAE, RMSE, and Spearman rank correlation. The composite score combined color, visibility, and translucency using weights of 0.4, 0.4, and 0.2, respectively.

The results supported the use of the VLM as a within-study ranking mechanism. However, the limited annotation set did not establish general calibration or transferability to other smoke datasets.

B. Background Preservation and Text-Image Alignment

Table XII summarizes background-preservation metrics and complete-image text–image alignment. FireSmokeGenNet obtained the highest point estimates under the common evaluation manifest. Because no uncertainty intervals were estimated for these image-level metrics, small differences between methods are interpreted descriptively.

C. Distribution Alignment

Table XIII reports the CLIP ViT-H/14 feature-space comparison. FireSmokeGenNet obtains the lowest distance under both reported distribution measures. These single-value comparisons are descriptive.

D. Boundary Softness Analysis

Table XIV reports the boundary-quality values. The value for FireSmokeGenNet is closest to the real-smoke reference among the compared methods, but it is not identical to the real-smoke distribution. Because the score is based on local gradient magnitude, it can also be affected by naturally low-contrast backgrounds and is not interpreted as a complete measure of perceptual realism.

The boundary-softness score does not independently measure perceptual realism because it can be influenced by background texture, contrast, and artificial smoothing. It is therefore interpreted as a local gradient-distribution descriptor rather than evidence that the generated boundaries are perceptually equivalent to real smoke.

TABLE X
FIVE-SEED COMPARISON OF REAL-ONLY AND MIXED-DATA TRAINING. PERFORMANCE IS REPORTED AS AP_{50} (%).

Detector	Condition	s_1	s_2	s_3	s_4	s_5	Mean ± SD	95% CI	Δ	Test	p_{adj}	d_z
YOLOv6	Real only	70.8	71.2	70.5	71.0	70.9	70.88±0.26	[70.56, 71.20]	–	–	–	–
	Mixed data	73.0	74.6	72.4	74.1	73.3	73.48±0.88	[72.39, 74.57]	+2.60	Paired t	0.0022	4.14
YOLOv7	Real only	72.1	72.5	71.9	72.4	72.2	72.22±0.24	[71.92, 72.52]	–	–	–	–
	Mixed data	74.0	75.7	74.5	74.6	75.3	74.82±0.68	[73.98, 75.66]	+2.60	Paired t	0.0021	4.63
YOLOv8	Real only	74.0	74.4	73.8	74.2	74.5	74.18±0.29	[73.82, 74.54]	–	–	–	–
	Mixed data	76.8	76.5	77.0	76.7	77.8	76.96±0.50	[76.34, 77.58]	+2.78	Paired t	0.0019	5.59
YOLOv9	Real only	75.2	75.6	75.0	75.5	75.3	75.32±0.24	[75.02, 75.62]	–	–	–	–
	Mixed data	77.3	78.9	77.8	77.4	78.5	77.98±0.70	[77.11, 78.85]	+2.66	Paired t	0.0022	4.19
YOLOv10	Real only	76.0	76.5	75.8	76.3	76.2	76.16±0.27	[75.82, 76.50]	–	–	–	–
	Mixed data	78.6	78.3	79.0	78.4	79.5	78.76±0.49	[78.15, 79.37]	+2.60	Paired t	0.0022	3.94
YOLOv11	Real only	76.8	77.2	76.5	77.0	77.1	76.92±0.28	[76.58, 77.26]	–	–	–	–
	Mixed data	79.0	80.4	79.2	79.0	80.0	79.52±0.64	[78.72, 80.32]	+2.60	Paired t	0.0021	5.25
YOLOv12	Real only	77.3	77.8	77.1	77.6	77.5	77.46±0.27	[77.12, 77.80]	–	–	–	–
	Mixed data	79.9	80.0	80.4	79.7	80.7	80.14±0.40	[79.64, 80.64]	+2.68	Paired t	0.0021	4.84
YOLOv13	Real only	78.0	78.5	77.8	78.3	78.2	78.16±0.27	[77.82, 78.50]	–	–	–	–
	Mixed data	80.1	81.8	80.5	80.6	81.4	80.88±0.70	[80.01, 81.75]	+2.72	Paired t	0.0021	5.12

TABLE XI
FIVE-FOLD CROSS-VALIDATION OF THE TASK-SPECIFIC VLM. VALUES ARE MEAN ± SAMPLE STANDARD DEVIATION ACROSS FOLDS.

Quality dimension	MAE ↓	RMSE ↓	Spearman ρ ↑
Color fidelity	0.62 ± 0.08	0.79 ± 0.10	0.76 ± 0.06
Visibility	0.57 ± 0.07	0.73 ± 0.09	0.81 ± 0.05
Translucency	0.71 ± 0.10	0.91 ± 0.12	0.69 ± 0.08
Composite score	0.53 ± 0.06	0.68 ± 0.08	0.83 ± 0.04

TABLE XII
COMPARISON OF BACKGROUND PRESERVATION AND TEXT–IMAGE ALIGNMENT. PSNR, SSIM, AND 1-LPIPS WERE COMPUTED OVER THE PRESERVED BACKGROUND REGION, WHEREAS CLIP-SIM WAS COMPUTED OVER THE COMPLETE GENERATED IMAGE. BEST REPORTED VALUES ARE SHOWN IN BOLD.

Method	PSNR ↑	SSIM ↑	1-LPIPS ↑	CLIP-Sim ↑
Stable Diffusion [20]	25.10	0.78	0.87	0.251
PowerPaint [50]	25.60	0.80	0.89	0.253
SD-Inpainting	26.00	0.81	0.90	0.255
BLD [51]	26.30	0.83	0.91	0.254
FlameDiffuser [23]	27.10	0.85	0.91	0.256
FireSmokeGenNet (Ours)	**28.10**	**0.88**	**0.92**	**0.258**

TABLE XIII
DISTRIBUTION ALIGNMENT BETWEEN REAL AND SYNTHETIC SMOKE. LOWER IS BETTER. FEATURES FROM CLIP VIT-H/14.

Method	CLIP-FD↓	$\mathrm{MMD}^2_{\mathrm{linear}}$ ↓
GAN-based synthesis [16]	45.32	0.124
Vanilla SD-inpainting	38.47	0.098
FlameDiffuser [23]	32.15	0.081
FireSmokeGenNet (Ours)	**28.15**	**0.062**

TABLE XIV
BOUNDARY QUALITY METRICS. REAL SMOKE VALUES SHOWN FOR REFERENCE.

Method	Boundary Softness $\mathcal{S}$	KL Divergence ↓
Real smoke	0.119	-
SD-Inpainting	0.203	0.187
FlameDiffuser	0.167	0.112
FireSmokeGenNet	**0.124**	**0.041**

E. Downstream Detection Performance

Table X provides the individual seed results, mean AP_{50}, sample standard deviation, two-sided 95% confidence interval, Holm-adjusted p-value, and paired effect size for each detector. The mixed-data condition improved mean AP_{50} for all eight detector families. Absolute improvements ranged from 2.60 to 2.78 percentage points, with an average gain of 2.66 percentage points.

The largest improvement was observed for YOLOv8, whose mean AP_{50} increased from $74.18 \pm 0.29\%$ to $76.96 \pm 0.50\%$. For YOLOv13, mean AP_{50} increased from $78.16 \pm 0.27\%$ to $80.88 \pm 0.70\%$, corresponding to an absolute improvement of 2.72 percentage points. All eight paired comparisons remained statistically significant after Holm correction. Because only five matched runs were available, the results were interpreted jointly using confidence intervals, adjusted p-values, and paired effect sizes.

Figure 9 and Table XVIII provide the equal-budget comparison across the evaluated training-data sources. At the same 28,800-image augmentation budget, VLM-ranked FireSmokeGenNet obtained the highest observed five-seed mean AP_{50}; the comparison is interpreted under the matched datasets and training protocol described in Section III-C.

F. Domain Generalization Under Environmental Shifts

Table XV evaluates cross-domain robustness. Training with FireSmokeGenNet synthetic data produced higher observed

TABLE XV
GENERALIZATION UNDER MONITORING DOMAIN SHIFTS. MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER FIVE YOLOV13 RUNS. AP_{50} AND RECALL ARE REPORTED ON THE DECIMAL SCALE.

Train $\to$ Test	Real Only		Real + FireSmokeGenNet	
	AP_{50}	Rec.	AP_{50}	Rec.
Summer $\to$ Winter	0.412 ± 0.023	0.385 ± 0.021	0.587 ± 0.018	0.542 ± 0.019
Day $\to$ Dusk	0.385 ± 0.031	0.352 ± 0.027	0.542 ± 0.022	0.498 ± 0.020
Clear $\to$ Haze/Fog	0.338 ± 0.028	0.306 ± 0.024	0.496 ± 0.025	0.458 ± 0.021
Average	**0.378**	**0.348**	**0.542**	**0.499**

TABLE XVI
FIXED-SEED ABLATION OF ARCHITECTURAL COMPONENTS USING YOLOV13. DETECTION METRICS ARE REPORTED ON THE DECIMAL SCALE.

Configuration	AP_{50}	$AP_{50:95}$	Prec.	Rec.
No JCA (CA only)	0.758	0.412	0.781	0.702
No MRDL ($\omega = 0$)	0.771	0.428	0.795	0.715
ResNet-50 without ImageNet pretraining	0.735	0.389	0.748	0.681
Unfiltered candidate pool	0.748	0.412	0.761	0.698
Full FireSmokeGenNet	**0.829**	**0.531**	**0.847**	**0.763**

AP_{50} and recall across the three evaluated environmental-shift scenarios.

Synthetic augmentation improved the mean AP_{50} in all three reported shifts, with the largest relative change occurring in the clear-to-haze/fog comparison. Exact means and standard deviations are reported in Table XV and are not repeated here. These comparisons are interpreted descriptively; no claim of statistical significance is made beyond the reported five-run summaries.

G. Qualitative Comparison

Figure 6 compares FireSmokeGenNet with SD-2 Inpainting, ControlNet, and a baseline latent-diffusion method. For each test case, all methods use the same smoke-free background, target smoke mask, text prompt, and DDIM sampling configuration. The examples include mountain-valley, forest-canopy, and hazy-hillside scenes.

a) Component Ablation: Table XIX and Figure 8 provide the primary variance-aware component comparison. Table XVI is retained as a separate fixed-seed development diagnostic because it additionally evaluates ImageNet initialization and the unfiltered candidate pool. Its full-model value is a single-run observation and therefore differs from the five-seed mean in Table X; causal component interpretations are based principally on the multi-seed comparison.

b) MRDL Weight Sensitivity: Figure 7 reports the fixed-seed MRDL-weight sensitivity analysis. AP_{50} peaked at $\omega = 0.4$, with degradation on either side. The three evaluated values are shown directly in Figure 7 and are not repeated in the text.

c) Filtering Threshold Sensitivity: Table XVII reports a fixed-seed exploratory retention-threshold analysis. The 30% setting obtained the strongest result in this sweep. Because the retained dataset size changed with the threshold, the comparison reflects the combined influence of sample quality and quantity and should not be interpreted as an isolated causal estimate of filtering quality.

TABLE XVII
FIXED-SEED SENSITIVITY TO THE SYNTHETIC-SAMPLE RETENTION THRESHOLD USING YOLOV13. DATASET SIZE VARIES WITH THE THRESHOLD; THEREFORE, THE COMPARISON REFLECTS BOTH SAMPLE QUALITY AND QUANTITY.

Retention Threshold	Dataset Size	AP_{50}	Recall
10%	9,600	0.812	0.741
20%	19,200	0.821	0.752
30%	28,800	**0.829**	**0.763**
40%	38,400	0.818	0.751
50%	48,000	0.805	0.738
No filtering	96,000	0.748	0.698

d) Additional Validation Summary: Table XIX and Figure 8 present the multi-seed component and regularization results. Table XX controls the retained sample count across selection strategies, Table XXI reports the pre-training leakage audit, and Table XXII summarizes the measured computational characteristics.

H. Discussion

1) Why Does FireSmokeGenNet Work?: Three design considerations provide possible interpretations of the observed results.

a) P1: Decoupled Structural and Contextual Conditioning: By processing the mask and masked image through separate ResNet-18 and ResNet-50 branches, respectively, the framework maintains separate structural and contextual feature streams before fusion. The frozen ImageNet-pretrained encoders provide complementary features, whereas the hierarchical injection supplies them at the corresponding U-Net stages.

b) P2: Stochastic Boundary Regularization Addresses Annotation Uncertainty: The morphological perturbations employed by MRDL represent plausible variations around an annotated smoke contour. Training for local consistency reduces the sensitivity to a single rigid boundary; however, it does not constitute a physical smoke simulation. The boundary measurements in Table XIV support this local consistency interpretation without implying a physical smoke model.

c) P3: Hierarchical Injection of Structural and Contextual Features: Low-resolution layers receive global mask constraints indicating where smoke should appear, whereas higher-resolution layers receive finer image-context features. This design supplies coarse structural conditioning at lower-resolution stages and finer contextual information at higher-resolution stages. The JCA module at intermediate resolutions enables interaction between the two feature streams.

2) Comparison with Prior Work:

a) vs. GAN-based Methods: GAN-based smoke synthesis can be affected by limited mode coverage and training instability [16]. In the present generative evaluation, diffusion-based methods achieved stronger background preservation and feature distribution alignment results than the evaluated GAN-based synthesis baseline. The equal-budget downstream comparison in Table XVIII further controls the number of retained synthetic images across the evaluated CycleGAN, inpainting, FlameDiffuser, and FireSmokeGenNet conditions.

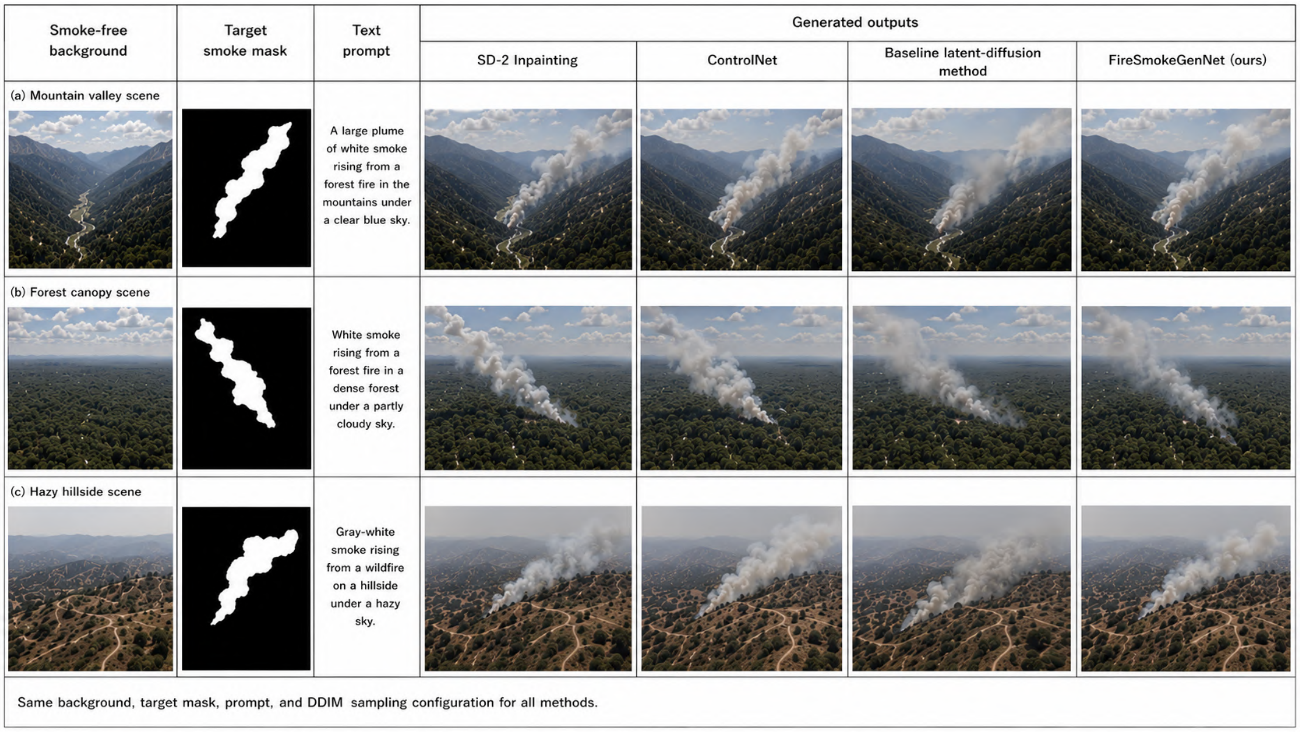

Fig. 6. **Qualitative comparison under matched generation conditions.** Each method uses the same smoke-free background, target smoke mask, text prompt, and DDIM sampling configuration for each test case.

TABLE XVIII
EQUAL-BUDGET DOWNSTREAM COMPARISON USING 28,800 SYNTHETIC TRAINING IMAGES FROM EACH GENERATOR. RESULTS ARE FORMATTED AS FIVE MATCHED YOLOV13 RUNS.

Training source	AP_{50} (%)	$AP_{50:95}$ (%)	Precision (%)	Recall (%)	95% CI for AP_{50}
Real only	78.16 ± 0.27	48.60 ± 0.35	81.20 ± 0.42	72.40 ± 0.51	$[77.82, 78.50]$
Real + GAN	79.02 ± 0.45	50.20 ± 0.48	82.10 ± 0.50	73.90 ± 0.62	$[78.46, 79.58]$
Real + SD-Inpainting	79.46 ± 0.52	50.87 ± 0.55	82.55 ± 0.48	74.62 ± 0.58	$[78.81, 80.11]$
Real + FlameDiffuser	80.03 ± 0.48	51.64 ± 0.51	83.16 ± 0.45	75.31 ± 0.55	$[79.43, 80.63]$
Real + FireSmokeGenNet, random	80.21 ± 0.61	52.12 ± 0.63	83.58 ± 0.52	75.74 ± 0.66	$[79.45, 80.97]$
Real + FireSmokeGenNet, VLM-ranked	80.88 ± 0.70	53.10 ± 0.68	84.70 ± 0.57	76.30 ± 0.71	$[80.01, 81.75]$

b) vs. Standard Diffusion Inpainting: Flat concatenation [20], [23] required the network to disentangle noisy-latent, mask, and image-context signals within a shared conditioning pathway. The proposed dual-branch design preserves separate structural and contextual representations before their interaction through the JCA. Under the reported evaluation, the complete system achieved stronger background preservation, boundary quality, and domain-shift results than the evaluated baselines; these comparisons are not interpreted as isolating a single causal mechanism.

c) vs. Simulation-Based Methods: Physics-based smoke-simulation approaches [14] commonly require numerical fluid modelling and scene-specific parameter selection, which can increase computational and configuration costs. The present diffusion-based approach instead learns an image distribution from observations and does not explicitly simulate fluid dynamics.

3) Limitations and Failure Modes: Despite its performance relative to the evaluated baselines, FireSmokeGenNet exhibits several limitations:

a) L1: Fog-Smoke Confusion: Qualitative inspection identified fog-smoke confusion as a recurring failure mode. Because no population-frequency estimate is claimed, this observation is reported as a limitation rather than a quantified prevalence. Possible mitigation includes explicit weather conditioning or contrastive separation.

b) L2: Over-Opacity in High-Contrast Scenes: Over-opacity in faint-smoke scenes was also observed qualitatively. No population-frequency estimate is claimed. Possible mitigation includes additional faint-smoke examples or density prediction as an auxiliary task.

c) L3: Limited Quality-Annotation Set: The VLM was evaluated using five-fold cross-validation on 150 human-rated images. Although the results supported within-study ranking, the limited sample size and absence of external validation constrained conclusions about calibration and transferability.

d) L4: Computational Cost: With 50 DDIM sampling steps, FireSmokeGenNet requires approximately 2 seconds per

TABLE XIX
FIVE-SEED COMPONENT AND REGULARIZATION COMPARISON USING YOLOV13 UNDER IDENTICAL PARTITIONS AND TRAINING SETTINGS.

Configuration	AP_{50} (%)	$AP_{50:95}$ (%)	Precision (%)	Recall (%)	ΔAP_{50}
Full FireSmokeGenNet	80.88 ± 0.70	53.10 ± 0.68	84.70 ± 0.57	76.30 ± 0.71	–
Without MRDL	77.18 ± 0.74	42.95 ± 0.71	79.62 ± 0.69	71.83 ± 0.77	-3.70
Standard consistency loss	78.36 ± 0.68	46.72 ± 0.65	81.08 ± 0.61	72.96 ± 0.70	-2.52
Boundary-weighted loss	79.04 ± 0.66	48.85 ± 0.64	82.14 ± 0.58	73.88 ± 0.68	-1.84
Gradient loss	78.71 ± 0.72	47.96 ± 0.70	81.72 ± 0.64	73.42 ± 0.73	-2.17
Dilation-only MRDL	79.46 ± 0.69	50.12 ± 0.66	82.83 ± 0.60	74.56 ± 0.71	-1.42
Erosion-only MRDL	79.29 ± 0.71	49.76 ± 0.69	82.51 ± 0.63	74.31 ± 0.72	-1.59
Conventional cross-attention	76.02 ± 0.79	41.38 ± 0.76	78.35 ± 0.71	70.44 ± 0.81	-4.86
Concatenation fusion	75.61 ± 0.82	40.64 ± 0.80	77.92 ± 0.75	69.86 ± 0.84	-5.27

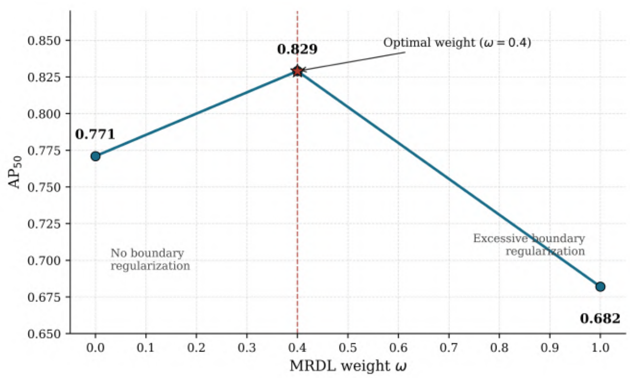

Fig. 7. **MRDL-weight sensitivity analysis.** AP_{50} reached its highest value at $\omega = 0.4$. Removing MRDL ($\omega = 0$) reduced AP_{50} to 0.771, whereas excessive weighting ($\omega = 1.0$) reduced AP_{50} to 0.682. Lines connect the three evaluated settings only as a visual guide.

TABLE XX
EQUAL-SIZE SYNTHETIC-SAMPLE SELECTION COMPARISON; EACH CONDITION CONTRIBUTES 28,800 IMAGES.

Selection strategy	AP_{50}	Recall
Random selection	80.21 ± 0.61	75.74 ± 0.66
Bottom-ranked by VLM	78.92 ± 0.73	72.81 ± 0.78
Rule-based filtering	80.02 ± 0.64	75.36 ± 0.69
Top-ranked by VLM	80.88 ± 0.70	76.30 ± 0.71

512×512 image on an A100 GPU. While acceptable for offline dataset generation, this is too slow for real-time applications. A potential direction is to reduce inference latency by distilling the model into a few-step sampler using progressive distillation [52] or by adopting a consistency-model formulation [53].

e) L5: Evaluation Completeness: The generative metrics were computed on fixed common manifests, but image-level confidence intervals and a systematic, blinded qualitative comparison were not included. Consequently, small metric differences were interpreted descriptively, and the evidence did not replace direct human comparison of smoke translucency, plume morphology, and boundary integration. The unfiltered comparison also changed the number of retained training samples and was therefore interpreted as a combined quality-quantity sensitivity analysis rather than as an isolated causal estimate of filtering.

TABLE XXI
DATASET LEAKAGE-AUDIT SUMMARY.

Audit category	Detected	Removed
Exact duplicates	184	184
Near-duplicate images	627	627
Shared source scenes/videos	0	0
Shared background identifiers	0	0
Shared mask-source identifiers	0	0

TABLE XXII
COMPUTATIONAL CHARACTERISTICS AT 512×512 RESOLUTION.

Method	Parameters	FLOPs	Peak memory	Time/image
SD-Inpainting	865 M	210 G	14.8 GB	1.6 s
FlameDiffuser	910 M	238 G	17.2 GB	1.9 s
FireSmokeGenNet	1.05 B	286 G	22.6 GB	2.0 s

4) Future Directions: Future work will extend FireSmokeGenNet toward temporally consistent latent-video smoke synthesis, physics-informed guidance, closed-loop active augmentation driven by detector uncertainty, and lightweight deployment-specific adapters. These directions target dynamic plume behaviour, stronger physical plausibility, more efficient sample selection, and lower adaptation cost in operational monitoring settings.

V. CONCLUSION

We presented FireSmokeGenNet, a latent-diffusion framework combining separate structural and contextual conditioning, joint cross-attention, boundary-restricted stochastic consistency regularization, and task-specific candidate ranking for wildfire-smoke synthesis. Under the reported evaluation protocol, adding 28,800 quality-filtered synthetic images to the real training set increased mean YOLOv13 AP_{50} from $78.16 \pm 0.27\%$ to $80.88 \pm 0.70\%$ across five matched seeds, corresponding to an absolute increase of 2.72 percentage points. The mixed-data mean had a 95% confidence interval of $[80.01, 81.75]\%$.

The background-region evaluation produced a PSNR of 28.10 dB and an SSIM of 0.88; these metrics measure preservation outside the conditioning smoke mask and should not be interpreted as direct measures of smoke realism. In the

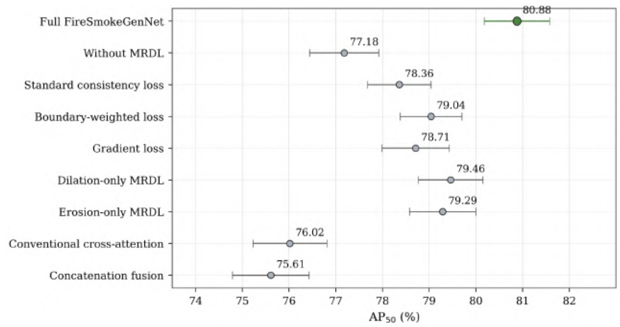

Fig. 8. Five-seed component and regularization comparison using YOLOv13. Points show mean AP_{50} and error bars show one sample standard deviation. The horizontal axis is truncated to emphasize differences; exact values are reported in Table XIX.

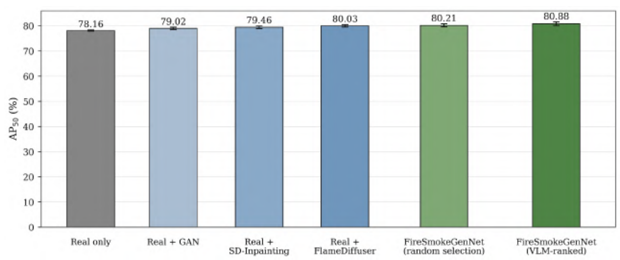

Fig. 9. Equal-budget downstream comparison using 28,800 synthetic images per augmentation condition. Bars show five-seed mean AP_{50} and error bars show one sample standard deviation.

five-seed comparison, the full system obtained higher observed AP_{50} than the evaluated loss and fusion alternatives. The separate fixed-seed development study also produced lower AP_{50} after removing ImageNet initialization or using the unfiltered candidate pool; those single-run findings remain exploratory. At an equal retained-image count, top-ranked VLM selection obtained a higher observed mean than random, bottom-ranked, and rule-based selection. The variable-size threshold sweep is interpreted separately because it changes both sample quality and quantity.

The findings indicate that boundary-aware diffusion augmentation can complement real wildfire-smoke training data under the evaluated datasets and detector configurations. Future work should examine external validation of quality ranking, other uncertain-boundary visual domains, independent monitoring networks, and faster sampling.

Ethics Statement

The human-rating component was limited to quality assessment of generated and real smoke images. Annotators were informed about the rating task before participation, responses were analyzed in aggregate, and no personal, medical, biometric, or sensitive participant attributes were used in model training or evaluation.

Data Availability

The public datasets used in this study are available from the original sources cited in the manuscript. Code, configurations, nonrestricted manifests, and derived result files will be released after repository validation and subject to third-party licensing conditions.

Disclosure of AI-Assisted Tools

Generative artificial-intelligence tools assisted with language editing and organization. All technical content, equations, experiments, numerical results, citations, and final wording were reviewed and verified by the authors, who take full responsibility for the manuscript.

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References

[1] L. Giglio, L. Boschetti, D. P. Roy, M. L. Humber, and C. O. Justice, "The collection 6 MODIS burned area mapping algorithm and product," *Remote Sens Environ*, vol. 217, pp. 72–85, Aug. 2018.

[2] P. Jain, Q. E. Barber, S. W. Taylor, E. Whitman, D. Castellanos Acuna, Y. Boulanger, R. D. Chavardès, J. Chen, P. Englefield, M. Flannigan, M. P. Girardin, C. C. Hanes, J. Little, K. Morrison, R. S. Skakun, D. K. Thompson, X. Wang, and M.-A. Parisien, "Drivers and impacts of the record-breaking 2023 wildfire season in canada," *Nature Communications*, vol. 15, no. 1, p. 6764, Aug 2024. [Online]. Available: https://doi.org/10.1038/s41467-024-51154-7

[3] B. Byrne, J. Liu, K. W. Bowman, M. Pascolini-Campbell, A. Chatterjee, S. Pandey, K. Miyazaki, G. R. van der Werf, D. Wunch, P. O. Wennberg, C. M. Roehl, and S. Sinha, "Carbon emissions from the 2023 canadian wildfires," *Nature*, vol. 633, no. 8031, pp. 835–839, Sep 2024. [Online]. Available: https://doi.org/10.1038/s41586-024-07878-z

[4] C. Tao, J. Zhang, and P. Wang, "Smoke detection based on deep convolutional neural networks," in *2016 International Conference on Industrial Informatics - Computing Technology, Intelligent Technology, Industrial Information Integration (ICIICII)*, 2016, pp. 150–153.

[5] A. Bouguettaya, H. Zarzour, A. M. Taberkit, and A. Kechida, "A review on early wildfire detection from unmanned aerial vehicles using deep learning-based computer vision algorithms," *Signal Processing*, vol. 190, p. 108309, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0165168421003467

[6] E. Karger, A. Jeppe, R. Ziolkowski, F. Korn, T. Hackel, M. Harr, T. Brée, and F. Ahlemann, "Artificial intelligence for wildfire detection and management," *Discover Artificial Intelligence*, vol. 6, no. 1, p. 304, Mar 2026. [Online]. Available: https://doi.org/10.1007/s44163-026-01087-5

[7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," 2016. [Online]. Available: https://arxiv.org/abs/1506.02640

[8] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," 2020. [Online]. Available: https://arxiv.org/abs/2005.12872

[9] M. Tan, R. Pang, and Q. V. Le, "Efficientdet: Scalable and efficient object detection," 2020. [Online]. Available: https://arxiv.org/abs/1911.09070

[10] A. Shamsoshoara, F. Afghah, A. Razi, L. Zheng, P. Z. Fulé, and E. Blasch, "Aerial imagery pile burn detection using deep learning: The flame dataset," *Computer Networks*, vol. 193, p. 108001, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1389128621001201

[11] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, and J. Gilmer, "The many faces of robustness: A critical analysis of out-of-distribution generalization," 2021. [Online]. Available: https://arxiv.org/abs/2006.16241

[12] L. B. Wanner, F. Lehner, L. Ziesel, J. Kuester, W. Gross, and A. Michel, "High quality synthetic particle data: simulating dust, smoke, and fog in Unreal Engine," in *Earth Resources and Environmental Remote Sensing/GIS Applications XVI*, K. Schulz, U. Michel, K. G. Nikolakopoulos, V. Gagliardi, and A. C. M. Teodoro, Eds., vol. 13671, International Society for Optics and Photonics. SPIE, 2025, p. 136710V. [Online]. Available: https://doi.org/10.1117/12.3070110

[13] D. Wang, E. Pajouheshgar, Y. Xu, T. Zhang, and S. Süsstrunk, "Volumetric temporal texture synthesis for smoke stylization using neural cellular automata," 2025. [Online]. Available: https://arxiv.org/abs/2502.09631

[14] A. Genovese, R. D. Labati, V. Piuri, and F. Scotti, "Virtual environment for synthetic smoke clouds generation," in *2011 IEEE International Conference on Virtual Environments, Human-Computer Interfaces and Measurement Systems Proceedings*, 2011, pp. 1–6.

[15] R. Donida Labati, A. Genovese, V. Piuri, and F. Scotti, "Wildfire smoke detection using computational intelligence techniques enhanced with synthetic smoke plume generation," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 43, no. 4, pp. 1003–1012, 2013.

[16] M. Park, D. Q. Tran, D. Jung, and S. Park, "Wildfire-detection method using densenet and cyclegan data augmentation-based remote camera imagery," *Remote Sensing*, vol. 12, no. 22, 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/22/3715

[17] C. Xie and H. Tao, "Generating realistic smoke images with controllable smoke components," *IEEE Access*, vol. 8, pp. 201 418–201 427, 2020.

[18] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," 2020. [Online]. Available: https://arxiv.org/abs/2006.11239

[19] A. Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models," 2021. [Online]. Available: https://arxiv.org/abs/2102.09672

[20] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," 2022. [Online]. Available: https://arxiv.org/abs/2112.10752

[21] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," 2023. [Online]. Available: https://arxiv.org/abs/2302.05543

[22] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, and X. Qie, "T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models," 2023. [Online]. Available: https://arxiv.org/abs/2302.08453

[23] H. Wang, S. P. H. Boroujeni, X. Chen, A. Bastola, H. Li, W. Zhu, and A. Razi, "Flame diffuser: Wildfire image synthesis using mask guided diffusion," 2024. [Online]. Available: https://arxiv.org/abs/2403.03463

[24] G. Wu, Y. Shang, C. Xu, H. Song, C. Wang, and Q. Zhang, "Mfgdiffusion: Mask-guided smoke synthesis for enhanced forest fire detection," 2025. [Online]. Available: https://arxiv.org/abs/2507.11252

[25] M. Sajjadi, M. Javanmardi, and T. Tasdizen, "Regularization with stochastic transformations and perturbations for deep semi-supervised learning," 2016. [Online]. Available: https://arxiv.org/abs/1606.04586

[26] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin, "Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution," 2024. [Online]. Available: https://arxiv.org/abs/2409.12191

[27] J. Xu, X. Liu, Y. Wu, Y. Tong, Q. Li, M. Ding, J. Tang, and Y. Dong, "Imagereward: Learning and evaluating human preferences for text-to-image generation," 2023. [Online]. Available: https://arxiv.org/abs/2304.05977

[28] X. Wu, K. Sun, F. Zhu, R. Zhao, and H. Li, "Human preference score: Better aligning text-to-image models with human preference," 2023. [Online]. Available: https://arxiv.org/abs/2303.14420

[29] Y. Chen, D. C. Morton, and J. T. Randerson, "Remote sensing for wildfire monitoring: Insights into burned area, emissions, and fire dynamics," *One Earth*, vol. 7, no. 6, pp. 1022–1028, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590332224002574

[30] A. Dewangan, Y. Pande, H.-W. Braun, F. Vernon, I. Perez, I. Altintas, G. W. Cottrell, and M. H. Nguyen, "FIgLib & SmokeyNet: Dataset and deep learning model for real-time wildland fire smoke detection," *Remote Sensing*, vol. 14, no. 4, p. 1007, Feb. 2022. [Online]. Available: http://dx.doi.org/10.3390/rs14041007

[31] E. Hamdan, Y. Luo, B. U. Toreyin, E. Koyuncu, A. J. Watts, U. Gudukbay, and A. E. Cetin, "A large scale open-source image and video dataset for robust wildfire detection and classification," 2026. [Online]. Available: https://arxiv.org/abs/2606.10174

[32] H. Cheng, J. Yin, B. Chen, and Z. Yu, "Smoke 100k: A database for smoke detection," in *IEEE 8th Global Conference on Consumer Electronics (GCCE)*, 2019, pp. 596–597.

[33] S. Yan, J. Zhang, and N. Barnes, "Transmission-guided bayesian generative model for smoke segmentation," 2023. [Online]. Available: https://arxiv.org/abs/2303.00900

[34] S. S. Mohamed, J. A. K. J, Pragya, M. H. Mir, G. B. R, and T. A. Mir, "Enhancing building safety through machine learning-based smoke detection," in *2025 Fifth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, 2025, pp. 1–6.

[35] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, and L. V. Gool, "Repaint: Inpainting using denoising diffusion probabilistic models," 2022. [Online]. Available: https://arxiv.org/abs/2201.09865

[36] S. Xie, Z. Zhang, Z. Lin, T. Hinz, and K. Zhang, "Smartbrush: Text and shape guided object inpainting with diffusion model," 2022. [Online]. Available: https://arxiv.org/abs/2212.05034

[37] T. Brooks, A. Holynski, and A. A. Efros, "Instructpix2pix: Learning to follow image editing instructions," 2023. [Online]. Available: https://arxiv.org/abs/2211.09800

[38] M. Aziz, U. Rehman, M. U. Danish, and K. Grolinger, "Global-local image perceptual score (glips): Evaluating photorealistic quality of ai-generated images," 2024. [Online]. Available: https://arxiv.org/abs/2405.09426

[39] J. Ho and T. Salimans, "Classifier-free diffusion guidance," 2022. [Online]. Available: https://arxiv.org/abs/2207.12598

[40] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2021. [Online]. Available: https://arxiv.org/abs/2011.13456

[41] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," 2021. [Online]. Available: https://arxiv.org/abs/2106.09685

[42] High Performance Wireless Research and Education Network, "HPWREN: High performance wireless research and education network," University of California San Diego, 2026, accessed: 2026-08-23. [Online]. Available: https://hpwren.ucsd.edu/

[43] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, "Segment anything," 2023. [Online]. Available: https://arxiv.org/abs/2304.02643

[44] Y. Tian, Q. Ye, and D. Doermann, "Yolov12: Attention-centric real-time object detectors," 2025. [Online]. Available: https://arxiv.org/abs/2502.12524

[45] M. Lei, S. Li, Y. Wu, H. Hu, Y. Zhou, X. Zheng, G. Ding, S. Du, Z. Wu, and Y. Gao, "Yolov13: Real-time object detection with hypergraph-enhanced adaptive visual perception," 2025. [Online]. Available: https://arxiv.org/abs/2506.17733

[46] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2022. [Online]. Available: https://arxiv.org/abs/2010.02502

[47] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[48] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," 2018. [Online]. Available: https://arxiv.org/abs/1801.03924

[49] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," 2021. [Online]. Available: https://arxiv.org/abs/2103.00020

[50] J. Zhuang, Y. Zeng, W. Liu, C. Yuan, and K. Chen, "A task is worth one word: Learning with task prompts for high-quality versatile image inpainting," 2024. [Online]. Available: https://arxiv.org/abs/2312.03594

[51] O. Avrahami, O. Fried, and D. Lischinski, "Blended latent diffusion," *ACM Transactions on Graphics*, vol. 42, no. 4, p. 1–11, Jul. 2023. [Online]. Available: http://dx.doi.org/10.1145/3592450

[52] T. Salimans and J. Ho, "Progressive distillation for fast sampling of diffusion models," 2022. [Online]. Available: https://arxiv.org/abs/2202.00512

[53] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, "Consistency models," 2023. [Online]. Available: https://arxiv.org/abs/2303.01469